
IEA Annex 57 - Case studies buildings

Overview

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Uster, 17.February 2015

Impressum

Titel	IEA Annex 57 - Case studies buildings
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Version	494_IEA Annex 57_Case studies_v0.1.docx, 2/17/2015 1:42:00 PM

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1 Introduction

The following subchapters 2.1 to 2.13 describe the case studies in their inventory, the parameter and the results according to the documentation requirements of the Annex 57.

Tab. 1.1 shows the mapping of the original buildings and the names anonymized.

Tab. 1.1 Original name of case study buildings and name anonymized

Name original	Name anonymized
Schulhaus Bläsi	School A
Schulhaus Ilgen A	School B
Schulhaus Ilgen B	School C
Schulhaus Holderbach	School D
Schulhaus Milchbuck	School E
Schulhaus Leutschenbach	School F
Schulhaus ZüriModular	School G
Wohngebäude Kehlhof	Residential building A
Wohngebäude Paradies Haus B	Residential building B
Wohngebäude Rautistrasse	Residential building E
Altersheim Dorflinde	Retirement home A
Pflegezentrum Witikon	Retirement home B
Altersheim Trotte	Retirement home C

2 Inventories and results per building

2.1 School A



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building (renovated school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school A has a total env. impact of 25'000 eco-points/m²a, primary energy demand (non renew.) of 420 MJ/m²a and a GWP of 11 kg CO₂/m²a. The building meets the target values for refurbished schools regarding global warming potential but not for primary energy demand (non renew). For all indicators the operation phase is dominating the results.

The roof, windows, flooring and the infrastructure cause the main impact within the construction stage (for CED and GWP) whereas the room heating (from electric heat pump equipped with a borehole heat exchanger) causes the largest impact in the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school A is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: School

Size: 2968 m² gross floor area, 2'606 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2015 (originally constructed in 1907)



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr nonrenewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School A

The school building A was originally built in 1907 and renovated in the 1970ties. Within the present renewal the façade, all technical installations and the interior structure will be replaced and refurbished. The refurbishment is combined with operational optimizations: the day care capability is extended from 80 to 100 positions.

The heat and hot water demand is covered by an electric heat pump equipped with a borehole heat exchanger while the heat distribution works with radiators.

CHARACTERISTIC FACTORS OPERATION

Floor area	2'968 m ²
Energy reference area	2'606 m ²
Energy demand room heating	238 MJ/m ² a
Energy demand hot water	10 MJm ² a
Energy demand electrical power	38 MJ/m ² a
- Energy demand ventilation:	0 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.74 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)

DETAILED RESULTS OF SCHOOL A, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school A in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0.06	3	3	-	-	0.0	0.0	0	-	-	0.00	0.00	0.00	-	-
	backfill	0	0	0	-	-	0.0	0.0	0.0	-	-	0.00	0.00	0.00	-	-
	fundament	18	1'101	823	-	279	0.3	15.1	14.6	-	0.5	0.02	1.06	0.72	-	0.34
	ceiling	92	5'526	4'857	-	669	0.6	34.0	29.1	-	4.9	0.06	3.78	3.51	-	0.27
	roof	385	23'089	11'345	5'672	6'072	4.8	287.3	187.4	93.7	6.1	0.30	18.02	9.99	5.00	3.03
	pillars	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls basement	23	1'375	1'248	-	127	0.2	14.6	13.8	-	0.9	0.03	1.87	1.83	-	0.04
	outer walls upper floors	41	2'445	2'319	-	127	0.6	35.1	34.3	-	0.8	0.06	3.47	3.44	-	0.03
	windows	764	45'866	18'959	18'959	7'948	8.5	510.4	252.6	252.6	5.2	0.56	33.73	15.55	15.55	2.63
	inner walls raw	95	5'722	4'980	-	742	0.9	52.3	47.0	-	5.4	0.08	5.06	4.77	-	0.28
	separation walls/inner doors	176	10'541	4'424	4'424	1'693	1.6	98.0	47.8	47.8	2.3	0.10	5.87	2.67	2.67	0.52
	flooring	885	53'125	20'734	20'734	11'657	7.7	463.8	227.5	227.5	8.7	0.53	32.07	14.22	14.22	3.63
	wall cover	210	12'628	5'358	5'358	1'911	1.9	114.6	54.5	54.5	5.7	0.18	10.76	4.64	4.64	1.47
	ceiling cover	575	34'489	16'509	16'509	1'470	4.0	242.1	118.6	118.6	4.9	0.26	15.60	7.19	7.19	1.23
	infrastructure	3'261	195'655	95'908	92'658	7'089	24.4	1'466.5	765.0	688.5	13.0	1.49	89.35	41.76	37.66	9.93
	sum building	6'526	391'565	187'467	164'315	39'783	55.6	3'334.0	1'792.3	1'483.2	58.4	3.68	220.63	110.30	86.93	23.41
Operation	room heating	9'489	569'339				196.0	11'757.1				3.07	183.95			
	hot water	631	37'878				13.2	790.4				0.21	12.37			
	ventilation	-	-				-	-				-	-			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	14'876	892'583				309.2	18'554.3				4.84	290.29			
Building induced mobility	sum mobility	3'224	193'430				49.6	2'975.4				2.65	158.85			
sum total	construction, operation und building induced mobility	24'626					414.4					11.16				
target							350.0					13.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school A per m² energy reference area and 60 years lifespan.

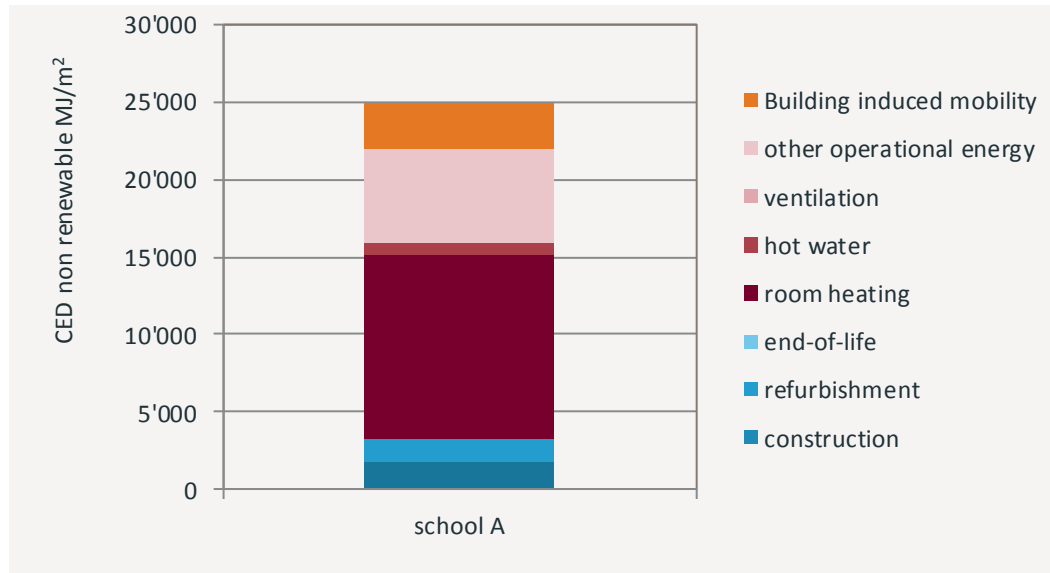


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school A

Total construction (construction, renewal and deconstruction): The total construction has a share of 13 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 54 % to this. The main impacts come from the floor and ceiling covers, windows, the roof, as well as the infrastructure. The renewal phase contributes to 44% to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 2 %.

The Swiss reference value regarding primary energy demand (non renew.) is not exceeded.

Operation: With 75 % the total operation has the main impact on the primary energy demand. The room heating has a share of 63% and the residual electricity demand causes 32% of the impacts. The Swiss reference level for refurbished school buildings is exceeded by 34 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility lies 17 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished building is about 18 % above than the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school A per m² energy reference area and 60 years lifespan.

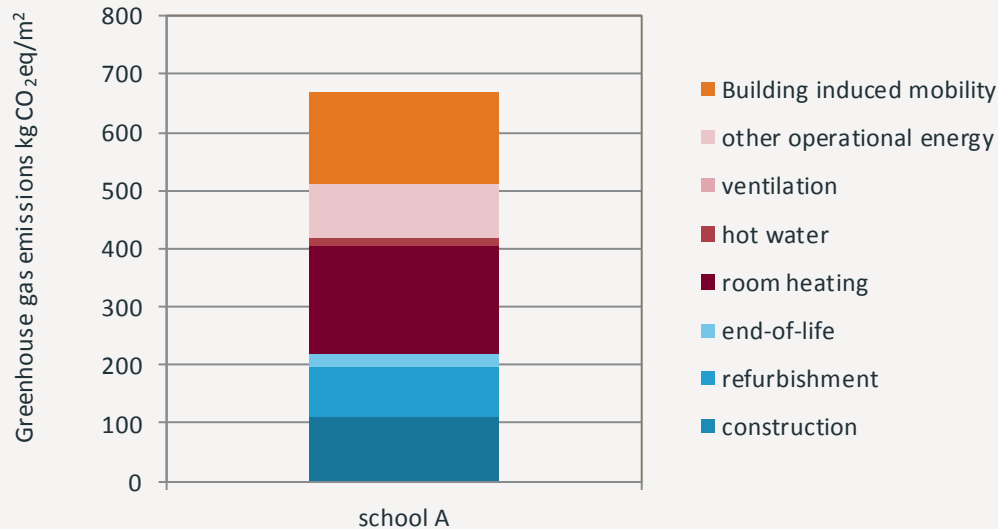


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school A.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 33 %. Within the construction phase the materialization is dominating the GWP (50%). The main impacts come from the floor and ceiling covers, windows, the roof, as well as the infrastructure. The renewal per year contributes about two fifth to the GWP of the total construction. The impact of the deconstruction has a share of 11 %. The reference level regarding GWP for refurbished schools is not exceeded.

Operation: In the building's use phase the GWP (43 %) is mainly influenced by the room heating and the residual electricity demand (63%, resp. 32%). The property school A comes with a heat pump. The reference level is not exceeded.

Induced mobility: The global warming potential of the induced mobility lies 12 % below the reference value.

Conclusion: The impacts of the use phase on the GWP are highest. The room heating and the electricity demand have the main impacts. Overall the GWP of the building school A is almost holds the target level for refurbished school buildings.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School A, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	Gross floor area 2968.06 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 2'606 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete/brick construction)
Thermal insulation	Exterior insulation of walls, roof insulation
Ventilation system	Ventilation of sanitary modules
Heating and cooling system	Heating: Bore hole heat exchanger, heat pump, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 0 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 238 MJ/m ² a (per energy reference area) Hot water 10 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie standard
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Building pit Foundation plate Ceilings Roof External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

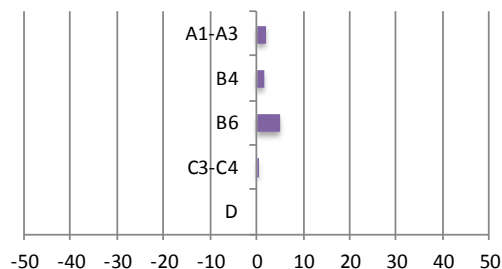
MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH1	School A	

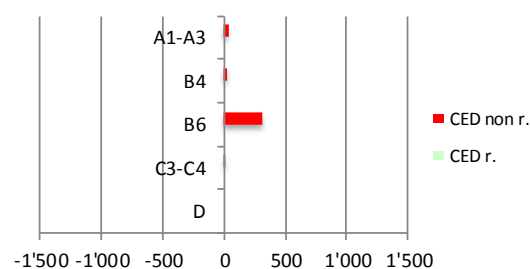
Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	1.84E+00	1.45E+00	4.84E+00	3.90E-01	0.00E+00
CED non r.	MJ/m ² a	2.99E+01	2.47E+01	3.09E+02	9.74E-01	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)

GWP 100a [kg CO₂ eq/m²a]



CED [MJ/m²a]



Inventories and results per building

2.2 School B



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building (renovated school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school B has a total env. impact of 23'000 eco-points/m²a, primary energy demand (non renew.) of 400 MJ/m²a and a GWP of 19 kg CO₂/m²a. The building does not meet the target values for refurbished schools regarding global warming potential nor for primary energy demand (non renew). For all indicators the operation phase is dominating the results.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school B is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: School

Size: 2'419 m² gross floor area, 1'759 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2013 (originally constructed in 1877)



ABBREVIATIONS

CED cumulative energy demand
GHG greenhouse gases
GWP global warming potential
LCA life cycle assessment
nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
- Frischknecht R., Steiner R. and Jungbluth N. (2008) Methode der ökologischen Knappheit - Ökofaktoren 2006. Umwelt-Wissen Nr. 0906. Bundesamt für Umwelt (BAFU), Bern, retrieved from: www.bafu.admin.ch/publikationen/publikation/01031/index.html?lang=de.
- Frischknecht R., Jungbluth N., Althaus H.-J., Bauer C., Doka G., Dones R., Hellweg S., Hischer R., Humbert S., Margni M. and Nemecek T. (2007) Implementation of Life Cycle Impact Assessment Methods. ecoinvent report No. 3, v2.0. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
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- SIA D 0236 (2011), SIA-Effizienzpfad Energie – Ergänzungen und Fallbeispiele zum Merkblatt SIA 2040

Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School B

The school building B was originally built in 1877. Within the present renewal the façade (wall insulation), all technical installations and the interior structure will be replaced and refurbished. The building reaches the label Minergie. Windows are automatized for ventilation. The heat and hot water demand is covered by district heat. The heat distribution works with radiators.

CHARACTERISTIC FACTORS OPERATION

Floor area	2'419 m ²
Energy reference area	1'759 m ²
Energy demand room heating	208 MJ/m ² a
Energy demand hot water	10 MJm ² a
Energy demand electrical power	38 MJ/m ² a
- Energy demand ventilation:	0 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.74 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)



DETAILED RESULTS OF SCHOOL B, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school B in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ²	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0.04	2	2	-	-	0.0	0.0	0.0	-	-	0.00	0.00	0.00	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	19	1'158	1'008	-	151	0.1	7.3	6.1	-	1.1	0.01	0.63	0.57	-	0.06
	ceiling	47	2'819	2'479	-	340	0.3	17.3	14.8	-	2.5	0.03	1.93	1.79	-	0.14
	roof	550	33'007	20'792	10'396	1'820	7.9	471.4	305.1	152.5	13.8	0.67	39.97	26.20	13.10	0.68
	pillars	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls basement	37	2'204	1'258	588	358	0.3	20.0	11.4	7.5	1.1	0.04	2.69	1.40	0.91	0.38
	outer walls upper floors	332	19'892	9'094	9'094	1'704	3.4	205.2	97.2	97.2	10.8	0.40	24.04	11.77	11.77	0.49
	windows	734	44'052	17'974	17'974	8'104	7.9	473.8	234.2	234.2	5.5	0.52	31.22	14.28	14.28	2.67
	inner walls raw	152	9'121	8'096	-	1'025	1.0	58.7	51.5	-	7.2	0.10	6.03	5.62	-	0.40
	separation walls/inner doors	143	8'586	3'247	3'247	2'092	1.3	78.1	38.3	38.3	1.5	0.07	4.43	1.90	1.90	0.63
	flooring	582	34'950	8'045	8'045	18'860	4.5	268.8	73.5	73.5	121.9	0.27	16.01	4.76	4.76	6.50
	wall cover	176	10'534	4'112	4'112	2'311	1.4	82.6	40.3	40.3	2.0	0.10	6.19	2.24	2.24	1.71
	ceiling cover	780	46'803	22'420	22'420	1'962	6.0	362.1	177.2	177.2	7.8	0.39	23.58	11.04	11.04	1.51
	infrastructure	5'244	314'622	152'795	156'095	5'731	26.8	1'610.1	785.7	812.0	12.4	1.62	97.47	44.03	45.63	7.81
	sum building	8'796	527'750	251'322	231'971	44'457	60.9	3'655.5	1'835.4	1'632.7	187.5	4.24	254.19	125.59	105.63	22.96
Operation	room heating	5'414	324'833				173.6	10'417.4				9.72	583.27			
	hot water	380	22'825				12.2	732.0				0.68	40.98			
	ventilation	-	-				-	-				-	-			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	10'550	633'023				285.9	17'156.3				11.97	718.24			
Building induced mobility	sum mobility	3'224	193'430				49.6	2'975.4				2.65	158.85			
sum total	construction, operation und building induced mobility	22'570					396.5					18.85				
target							350.0					13.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school B per m² energy reference area and 60 years lifespan.

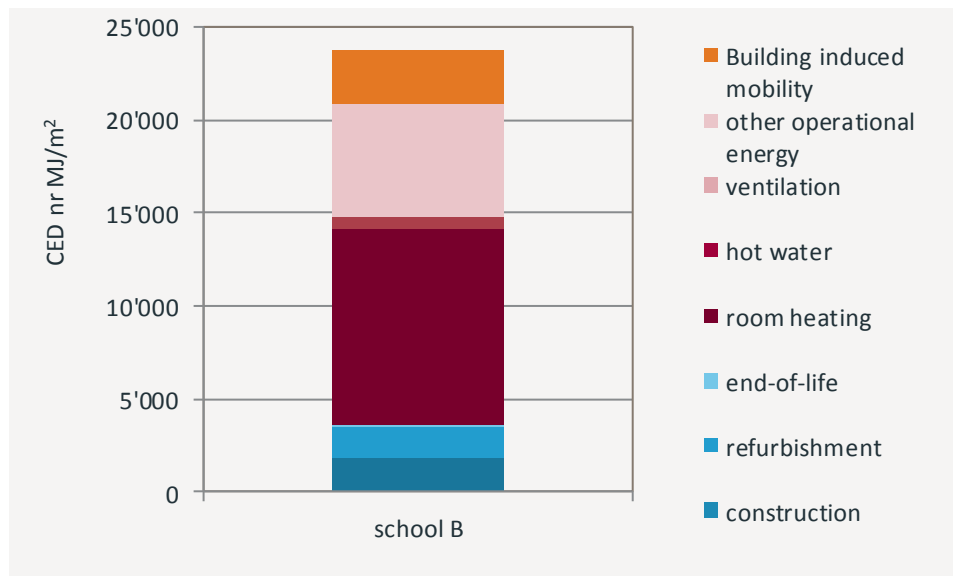


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school B.

Total construction (construction, renewal and deconstruction): The total construction has a share of 15 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 50 % to this. The main impacts come from the floor and ceiling covers, windows, the roof, as well as the infrastructure. The renewal phase contributes to 45 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 5 %.

The Swiss reference value regarding primary energy demand (non renew.) is slightly exceeded.

Operation: With 72 % the total operation has the main impact on the primary energy demand. The room heating has a share of 61% and the residual electricity demand causes 35 % of the impacts. The Swiss reference level for refurbished school buildings is exceeded by 24 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility lies 17 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished building is about 13 % above than the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school B per m² energy reference area and 60 years lifespan.

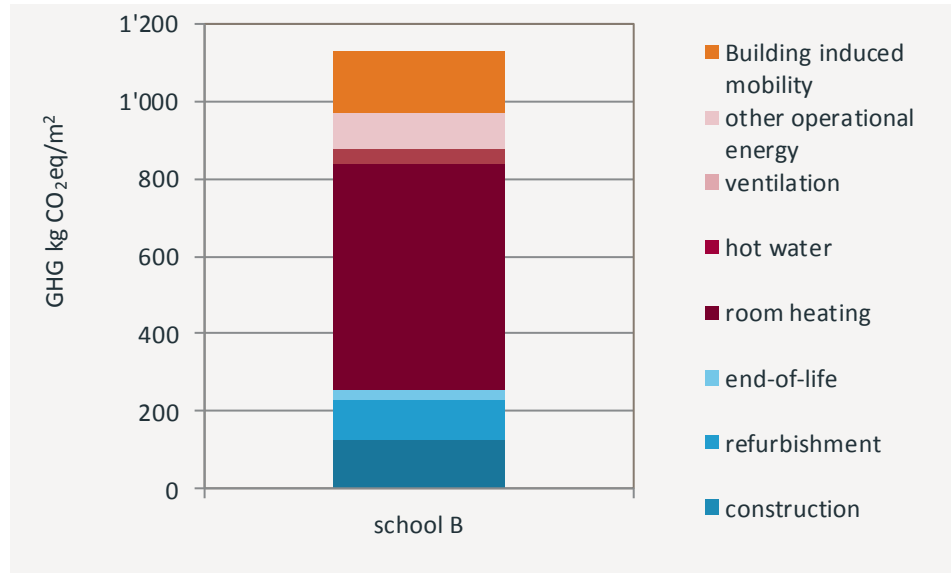


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school B.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 22 %. Within the construction phase the materialization is dominating the GWP (49 %). The main impacts come from the floor and ceiling covers, windows, the roof, as well as the infrastructure. The renewal per year contributes about 42 % to the GWP of the total construction. The deconstruction has a share of 9 %. The reference level regarding GWP for refurbished schools is not exceeded.

Operation: In the building's use phase the GWP is mainly influenced by the room heating (81 %). The property school B is heated with district heat. The reference level is exceeded by 140 %.

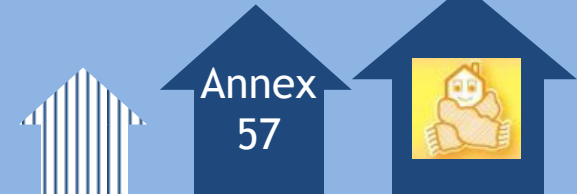
Induced mobility: The global warming potential of the induced mobility lies 12 % below the reference value.

Conclusion: The impacts of the operation phase on the GWP are highest. The room heating has the main impact. Overall the GWP of the building school B exceeds the target level for refurbished school buildings by 40 %.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School B, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	2'419 m ² / n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 1'759 m ²
Surface/Volume ratio (m ⁻¹)	n/a
Construction method	Massive construction (stone/concrete/brick construction)
Thermal insulation	Exterior insulation of walls, roof insulation
Ventilation system	Automatic window ventilation
Heating and cooling system	Heating: district heat, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 0 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 208 MJ/m ² a (per energy reference area) Hot water 10 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie standard
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Building pit Foundation plate Ceilings Roof External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	-
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

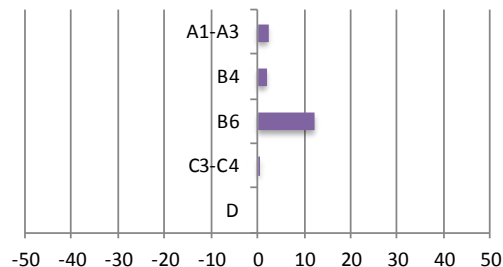
MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		
CH2	School B	57 Annex 57

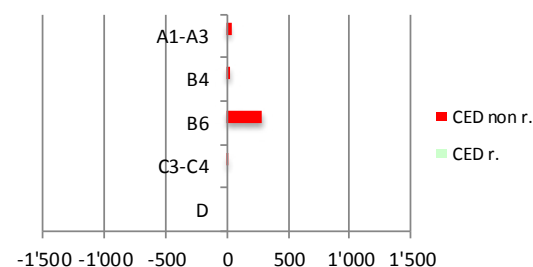
Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	2.09E+00	1.76E+00	1.20E+01	3.83E-01	0.00E+00
CED non r.	MJ/m ² a	3.06E+01	2.72E+01	2.86E+02	3.12E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)

GWP 100a [kg CO₂ eq/m²a]



CED [MJ/m²a]



Project: Richtwert Gesamtumweltbelastung Gebäude
 Carried out by: treeze Ltd, Architekturbüro Preisig Pfäffli, ETH Zürich
 Database: KBOB, ecoinvent

Inventories and results per building

2.3 School C



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building (renovated school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school C has a total env. impact of 21'000 eco-points/m²a, primary energy demand (non renew.) of 360 MJ/m²a and a GWP of 17 kg CO₂/m²a. The building does not meet the target values for refurbished schools regarding global warming potential nor for primary energy demand (non renew). For all indicators the operation phase is dominating the results.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school C is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: School

Size: 3'560 m² gross floor area, 2'900 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2013 (originally constructed in 1877)



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
- Frischknecht R., Steiner R. and Jungbluth N. (2008) Methode der ökologischen Knappheit - Ökofaktoren 2006. Umwelt-Wissen Nr. 0906. Bundesamt für Umwelt (BAFU), Bern, retrieved from: www.bafu.admin.ch/publikationen/publikation/01031/index.html?lang=de.
- Frischknecht R., Jungbluth N., Althaus H.-J., Bauer C., Doka G., Dones R., Hellweg S., Hischer R., Humbert S., Margni M. and Nemecek T. (2007) Implementation of Life Cycle Impact Assessment Methods. ecoinvent report No. 3, v2.0. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- IPCC (2007) The IPCC fourth Assessment Report - Technical Summary. Cambridge University Press., Cambridge.
- ecoinvent Centre (2010) ecoinvent data v2.2, ecoinvent reports No. 1-25. Swiss Centre for Life Cycle Inventories, Dübendorf, Switzerland, retrieved from: www.ecoinvent.org.
- SIA D 0236 (2011), SIA-Effizienzpfad Energie – Ergänzungen und Fallbeispiele zum Merkblatt SIA 2040

Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School C

The school building C was originally built in 1877. Within the present renewal the façade (wall insulation), all technical installations and the interior structure will be replaced and refurbished. The building reaches the label Minergie. Windows are automatized for ventilation. The heat and hot water demand is covered by district heat. The heat distribution works with radiators.

CHARACTERISTIC FACTORS OPERATION

Floor area	3'560 m ²
Energy reference area	2'900 m ²
Energy demand room heating	158 MJ/m ² a
Energy demand hot water	20 MJm ² a
Energy demand electrical power	38 MJ/m ² a
- Energy demand ventilation:	0 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.74 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)



DETAILED RESULTS OF SCHOOL C, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school C in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ²	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0.02	1	1	-	-	0.0	0.0	0.0	-	-	0.00	0.00	0.00	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	12	703	611	-	91	0.1	4.4	3.7	-	0.7	0.01	0.38	0.34	-	0.04
	ceiling	41	2'487	2'187	-	300	0.3	15.3	13.1	-	2.2	0.03	1.70	1.58	-	0.12
	roof	483	28'991	18'294	9'147	1'549	6.9	413.7	267.7	133.9	12.2	0.59	35.26	23.11	11.56	0.59
	pillars	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls basement	11	640	397	143	99	0.1	5.5	3.3	1.8	0.3	0.01	0.73	0.41	0.22	0.10
	outer walls upper floors	319	19'143	8'848	8'704	1'592	3.4	202.1	97.2	94.8	10.1	0.38	22.99	11.38	11.15	0.46
	windows	576	34'562	14'099	14'099	6'363	6.2	371.9	183.8	183.8	4.3	0.41	24.50	11.20	11.20	2.09
	inner walls raw	99	5'956	5'261	-	695	0.7	40.4	35.5	-	4.9	0.07	4.11	3.84	-	0.27
	separation walls/inner doors	397	23'810	10'713	9'717	3'380	4.2	249.5	128.5	115.2	5.8	0.28	16.72	8.37	7.13	1.22
	flooring	256	15'359	6'440	6'440	2'479	1.9	116.7	57.7	57.7	1.2	0.12	7.28	3.45	3.45	0.38
	wall cover	162	9'715	3'851	3'851	2'013	1.3	77.8	37.9	37.9	2.1	0.10	5.78	2.11	2.11	1.57
	ceiling cover	302	18'117	8'439	8'439	1'238	2.4	142.2	69.6	69.6	3.1	0.16	9.56	4.21	4.21	1.14
	infrastructure	5'244	314'622	152'795	156'095	5'731	26.8	1'610.1	785.7	812.0	12.4	1.62	97.47	44.03	45.63	7.81
	sum building	7'902	474'106	231'938	216'637	25'532	54.2	3'249.6	1'683.7	1'506.6	59.3	3.77	226.48	114.02	96.67	15.79
Operation	room heating	4'112	246'744				131.9	7'913.1				7.38	443.06			
	hot water	761	45'650				24.4	1'464.0				1.37	81.97			
	ventilation	-	-				-	-				-	-			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	9'629	577'760				256.4	15'384.0				10.32	619.01			
Building induced mobility	sum mobility															
		3'224	193'430				49.6	2'975.4				2.65	158.85			
sum total	construction, operation und building induced mobility	20'755					360.1					16.74				
target							350.0					13.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school C per m² energy reference area and 60 years lifespan.

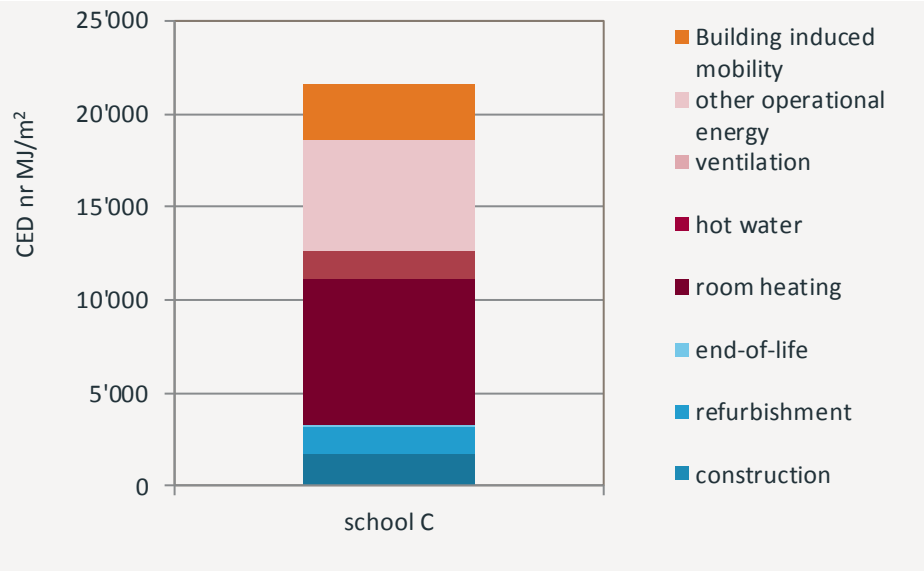


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school C.

Total construction (construction, renewal and deconstruction): The total construction has a share of 15 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 52 % to this. The main impacts come from the outer and inner walls, windows, the roof, as well as the infrastructure. The renewal phase contributes to 46 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 2 %.

The Swiss reference value regarding primary energy demand (non renew.) is slightly exceeded.

Operation: With 71 % the total operation has the main impact on the primary energy demand. The room heating has a share of 51 % and the residual electricity demand causes 39 % of the impacts. The Swiss reference level for refurbished school buildings is exceeded by 11 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility lies 29 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished building is about 3% above than the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school C per m² energy reference area and 60 years lifespan.

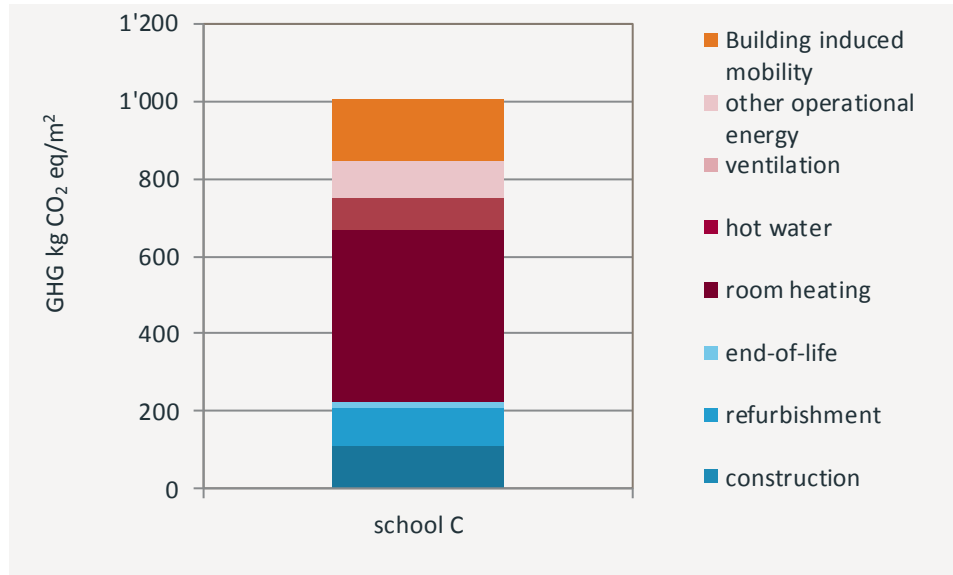


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school C.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 23 %. Within the construction phase the materialization is dominating the GWP (50 %). The main impacts come from the outer walls, windows, the roof, as well as the infrastructure. The renewal per year contributes about 43 % to the GWP of the total construction. The deconstruction has a share of 7 %. The reference level regarding GWP for refurbished schools is not exceeded.

Operation: In the building's use phase the GWP (62 %) is mainly influenced by the room heating and the electricity demand (72 % resp. 15 %). The property school C is heated with district heat. The reference level is exceeded by 106 %.

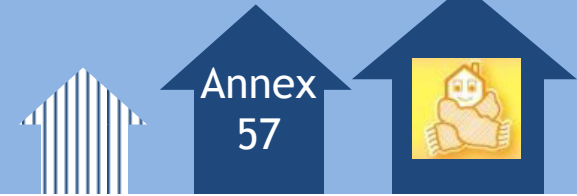
Induced mobility: The global warming potential of the induced mobility lies 12 % below the reference value.

Conclusion: The impacts of the operation phase on the GWP are highest. The room heating has the main impact. Overall the GWP of the building school C exceeds the target level for refurbished school buildings by 24 %.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School C, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	3'560 m ² / n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 2'900 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (stone/concrete/brick construction)
Thermal insulation	Exterior insulation of walls, roof insulation
Ventilation system	Automatic window ventilation
Heating and cooling system	Heating: district heat, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 0 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 158 MJ/m ² a (per energy reference area) Hot water 20 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie standard
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

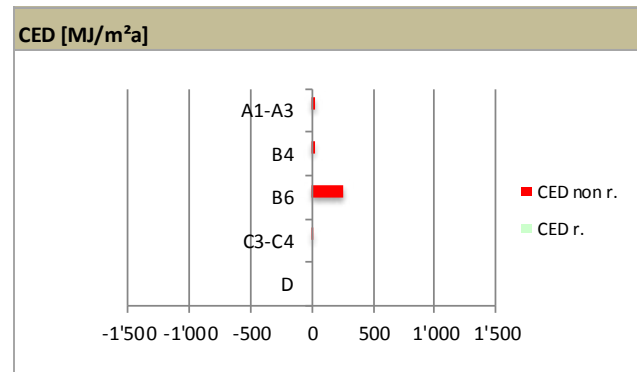
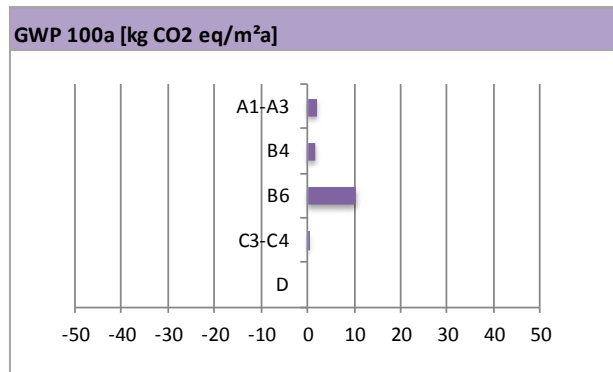
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Building pit Foundation plate Ceilings Roof External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH3	School C	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	1.90E+00	1.61E+00	1.03E+01	2.63E-01	0.00E+00
CED non r.	MJ/m ² a	2.81E+01	2.51E+01	2.56E+02	9.88E-01	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



Project: Richtwert Gesamtumweltbelastung Gebäude
 Carried out by: treeze Ltd, Architekturbüro Preisig Pfäffli, ETH Zürich
 Database: **KBOB**, **ecoinvent**

Inventories and results per building

2.4 School D



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building D (refurbished school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school D has a total env. impact of 34'000 eco-points/m²a, primary energy demand (non renew.) of 560 MJ/m²a and a GWP of 17 kg CO₂/m²a. The building does not meet the target values for refurbished schools regarding global warming potential nor for primary energy demand (non renew). For the indicators total environmental impact and CED nr the operation stage is dominating the results while the construction stage dominates the GHG emissions.

The roof, windows and the infrastructure cause the main impact within the construction stage (for CED and GWP) whereas the room heating (from electric heat pump equipped with a borehole heat exchanger) causes the largest impact in the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school D is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

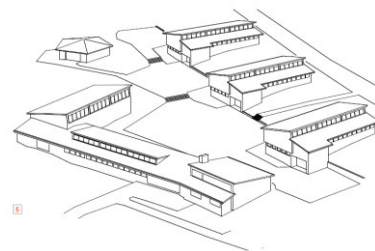
BUILDING KEY FACTS

Intended use: School

Size: Gross floor area n/a, 3'057 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2010 (originally constructed in 1950)



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

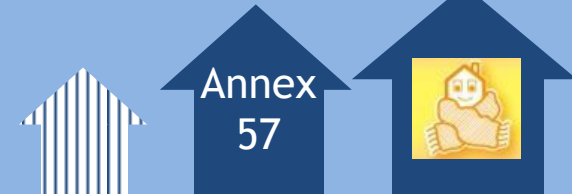
- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
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- SIA D 0236 (2011), SIA-Effizienzpfad Energie – Ergänzungen und Fallbeispiele zum Merkblatt SIA 2040

Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School D

The school building D was originally built in the 1950ies. It consists of three school pavilions and a kindergarten. Within the present renewal the façade (wall, roof and ceiling insulation), all technical installations and the interior structure will be refurbished. The building reaches the label Minergie. Sanitary rooms have automatized ventilation.

The heat and hot water demand is covered by electric heat pump equipped with a borehole heat exchanger. The heat distribution works with radiators.

CHARACTERISTIC FACTORS OPERATION

Floor area	n/a
Energy reference area	3'057 m ²
Energy demand room heating	300 MJ/m ² a
Energy demand hot water	10 MJm ² a
Energy demand electrical power	38 MJ/m ² a
- Energy demand ventilation:	0 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.74 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)



DETAILED RESULTS OF SCHOOL D, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school D in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	1.72	103	103	-	-	0.0	1.2	1.2	0.0	0.0	0.00	0.08	0.08	-	-
	backfill	2	93	93	-	-	0.0	1.1	1.1	0.0	0.0	0.00	0.08	0.08	-	-
	fundament	57	3'440	2'947	-	493	0.4	21.5	17.8	0.0	3.7	0.03	1.93	1.73	-	0.20
	ceiling	32	1'938	1'709	-	229	0.2	11.9	10.1	0.0	1.7	0.02	1.33	1.24	-	0.09
	roof	3'167	190'005	81'779	40'889	67'337	60.3	3'619.3	2'405.5	1'202.7	11.1	3.88	232.93	85.49	42.74	104.70
	pillars	93	5'556	5'556	-	-	0.5	32.2	32.2	0.0	0.0	0.03	1.91	1.91	-	-
	outer walls basement	8	490	432	-	58	0.0	3.0	2.6	0.0	0.4	0.01	0.34	0.31	-	0.02
	outer walls upper floors	339	20'358	8'198	8'198	3'961	3.8	229.8	112.4	112.4	5.0	0.50	30.01	12.34	12.34	5.33
	windows	1'324	79'455	33'668	33'668	12'118	15.3	918.9	455.5	455.5	7.9	1.06	63.59	28.73	28.73	6.12
	inner walls raw	19	1'165	1'008	-	157	0.3	17.8	16.8	0.0	1.1	0.03	1.56	1.49	-	0.07
	separation walls/inner doors	126	7'547	2'722	2'722	2'103	1.2	73.7	36.1	36.1	1.6	0.07	4.48	1.90	1.90	0.68
	flooring	483	28'987	11'603	11'603	5'780	6.5	391.8	194.1	194.1	3.6	0.36	21.36	8.59	8.59	4.18
	wall cover	248	14'864	6'856	6'856	1'153	2.0	121.5	58.6	58.6	4.2	0.12	6.99	3.09	3.09	0.81
	ceiling cover	917	55'031	25'802	25'802	3'428	7.5	452.3	221.5	221.5	9.4	0.43	25.52	11.81	11.81	1.90
	infrastructure	5'989	359'322	165'895	187'695	5'731	42.8	2'568.3	1'106.6	1'449.4	12.4	2.50	150.07	61.62	80.65	7.81
	sum building	12'806	768'355	348'373	317'434	102'548	141.1	8'464.5	4'672.1	3'730.4	62.0	9.04	542.18	220.41	189.86	131.91
Operation	room heating	12'336	740'154				255.0	15'297.7				3.99	239.34			
	hot water	631	37'878				13.2	790.4				0.21	12.37			
	ventilation	-	-				0.0	0.0				-	-			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	17'723	1'063'398				368.2	22'094.9				5.76	345.69			
Building induced mobility	sum mobility	3'224	193'430				49.6	2'975.4				2.65	158.85			
sum total	construction, operation und building induced mobility	33'753					558.9					17.45				
target							350.0					13.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school D per m² energy reference area and 60 years lifespan.

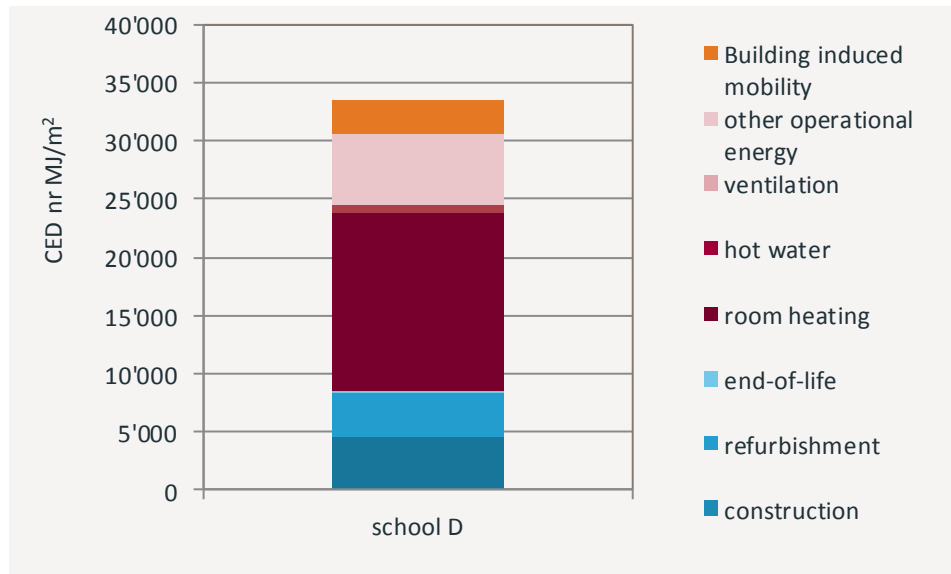


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school D.

Total construction (construction, renewal and deconstruction): The total construction has a share of 25 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 55 % to this. The main impacts come from the roof, windows, as well as the infrastructure. The renewal phase contributes to 44 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 1 %.

The Swiss reference value regarding primary energy demand (non renew.) is exceeded by 135 %.

Operation: With 66 % the total operation has the main impact on the primary energy demand. The room heating has a share of 69 % and the residual electricity demand causes 27 % of the operational impacts. The Swiss reference level for refurbished school buildings is exceeded by 60 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility lies 17 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished school building is about 60 % above than the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school D per m² energy reference area and 60 years lifespan.

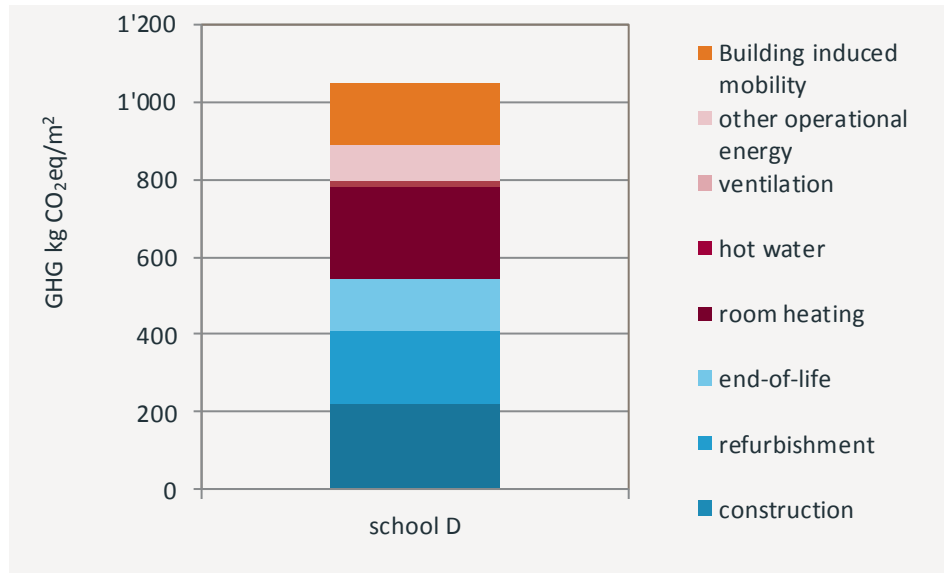


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school D.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 52 %. Within the construction phase the materialization is dominating the GWP (41 %). The main impacts come from the roof, the windows as well as the infrastructure. The renewal per year contributes about 35 % to the GWP of the total construction. The deconstruction has a share of 24 %. The reference level regarding GWP for refurbished schools is exceeded by 64 %.

Operation: In the building's use phase the GWP (33 %) is mainly influenced by the room heating and the electricity demand (69 % resp. 27 %). The property school D is heated with district heat. The reference level is exceeded by 15 %.

Induced mobility: The global warming potential of the induced mobility lies 12 % below the reference value.

Conclusion: The impacts of the construction phase on the GWP are highest. The roof and the infrastructure have the main impact. Overall the GWP of the building school D exceeds the target level for refurbished school buildings by 29 %.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School D, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 3'057 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (stone/concrete/brick construction)
Thermal insulation	Exterior insulation of walls, roof insulation
Ventilation system	Automatic window ventilation
Heating and cooling system	Heating:electric heat pump equipped with a borehole heat exchanger, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 0 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 300 MJ/m ² a (per energy reference area) Hot water 10 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie standard
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

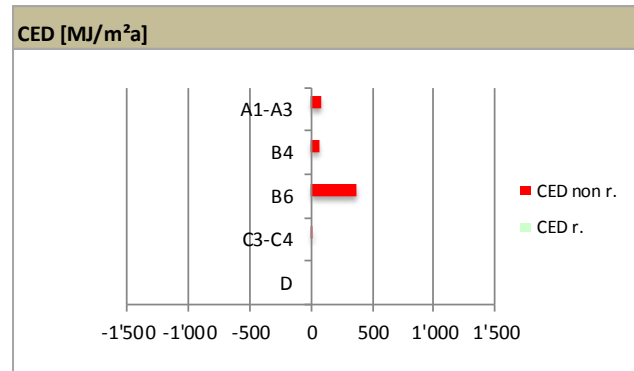
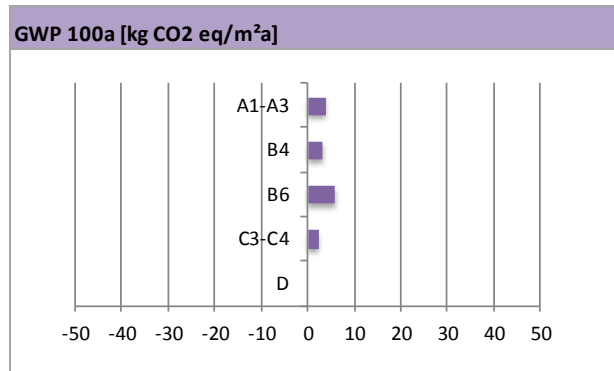
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials uses for the renewal were considered. Building pit Backfill Foundation plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH4	School D	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	3.67E+00	3.16E+00	5.76E+00	2.20E+00	0.00E+00
CED non r.	MJ/m ² a	7.79E+01	6.22E+01	3.68E+02	1.03E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



Inventories and results per building

2.5 School E



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building E (refurbished school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school E has a total env. impact of 31'000 eco-points/m²a, primary energy demand (non renew.) of 320 MJ/m²a and a GWP of 14 kg CO₂/m²a. The building holds the target values for refurbished schools regarding global warming potential and for primary energy demand (non renew). For all indicators the operation stage is dominating the results.

The roof, windows, flooring and the infrastructure cause the main impact within the construction stage (for CED and GWP) whereas the electricity demand (CED) and room heating (GWP, from wood pellet furnace) causes the largest impact in the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school E is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

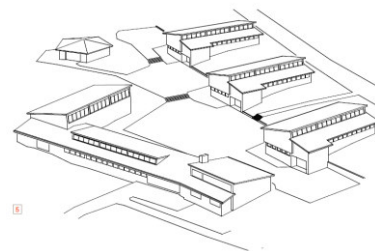
BUILDING KEY FACTS

Intended use: School

Size: 14'058 m² Gross floor area, 8'033 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2009 (originally constructed in 1930)



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

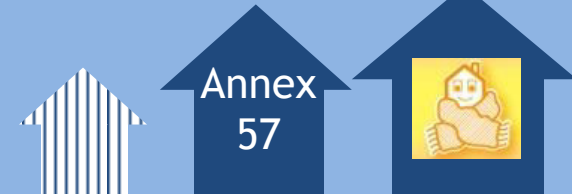
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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School E

The building E is a school property built in 1930. Only few modifications were made in the past. In 2009 the school was completely renovated and the building corresponds to the Swiss Minergie standard for refurbished buildings. The classroom wing was broadly renovated and the classrooms were enlarged.

The structural components are sandstone blocks, concrete and building bricks. The new windows have wooden frames.

The heat and hot water demand is covered by district heat and wood pellets and the heat is distributed by radiators. The building is equipped with an automatic ventilation.

CHARACTERISTIC FACTORS OPERATION

Floor area	14'058
Energy reference area	8'033 m ²
Energy demand room heating	239 MJ/m ² a
Energy demand hot water	10 MJm ² a
Energy demand electrical power	38 MJ/m ² a
- Energy demand ventilation:	0 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.20 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)

DETAILED RESULTS OF SCHOOL E, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school E in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0	14	14	-	-	0.0	0.2	0.2	0.0	0.0	0.00	0.01	0.01	-	-
	backfill	0	10	10	-	-	0.0	0.1	0.1	0.0	0.0	0.00	0.01	0.01	-	-
	fundament	7	405	348	-	57	0.0	2.5	2.1	0.0	0.4	0.00	0.23	0.20	-	0.02
	ceiling	377	22'634	19'965	-	2'669	2.3	138.6	118.3	0.0	20.3	0.26	15.55	14.46	-	1.09
	roof	668	40'103	16'933	16'933	6'236	3.8	225.1	109.0	109.0	7.0	0.26	15.74	6.62	6.62	2.50
	pillars	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	-	-	-	-	-
	outer walls basement	29	1'740	1'496	11	233	0.2	12.1	10.6	0.2	1.4	0.02	1.38	1.21	0.02	0.15
	outer walls upper floors	209	12'560	5'286	5'286	1'987	2.7	159.6	77.7	77.7	4.3	0.28	16.59	7.05	7.05	2.49
	windows	479	28'730	11'817	11'817	5'097	5.3	319.9	158.2	158.2	3.6	0.35	21.11	9.70	9.70	1.71
	inner walls raw	159	9'553	8'354	-	1'199	1.6	95.4	86.3	0.0	9.1	0.16	9.34	8.87	-	0.47
	separation walls/inner doors	186	11'176	4'511	4'511	2'154	1.8	106.4	51.7	51.7	2.9	0.11	6.77	3.04	3.04	0.68
	flooring	995	59'711	25'706	25'706	8'298	8.6	516.6	250.7	250.7	15.2	0.62	37.09	14.86	14.86	7.36
	wall cover	255	15'277	6'277	6'277	2'722	2.4	145.5	68.9	68.9	7.7	0.25	15.26	6.55	6.55	2.15
	ceiling cover	612	36'732	16'575	16'575	3'583	4.9	294.6	143.3	143.3	8.0	0.32	19.30	8.44	8.44	2.42
	infrastructure	5'228	313'671	151'220	156'720	5'731	28.1	1'686.6	815.1	859.1	12.4	1.70	101.92	45.72	48.39	7.81
	sum building	9'205	552'317	268'513	243'836	39'968	61.7	3'703.3	1'892.3	1'718.8	92.3	4.34	260.29	126.75	104.67	28.86
Operation	room heating	11'648	698'901				88.3	5'297.5				4.46	267.38			
	hot water	732	43'901				5.5	332.8				0.28	16.80			
	ventilation	-	-				0.0	0.0				-	-			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	17'136	1'028'167				194.0	11'637.2				6.30	378.16			
Building induced mobility	sum mobility	4'618	277'105				64.5	3'872.7				3.67	220.00			
sum total	construction, operation und building induced mobility	30'960					320.2					14.31				
target							350.0					13.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school E per m² energy reference area and 60 years lifespan.

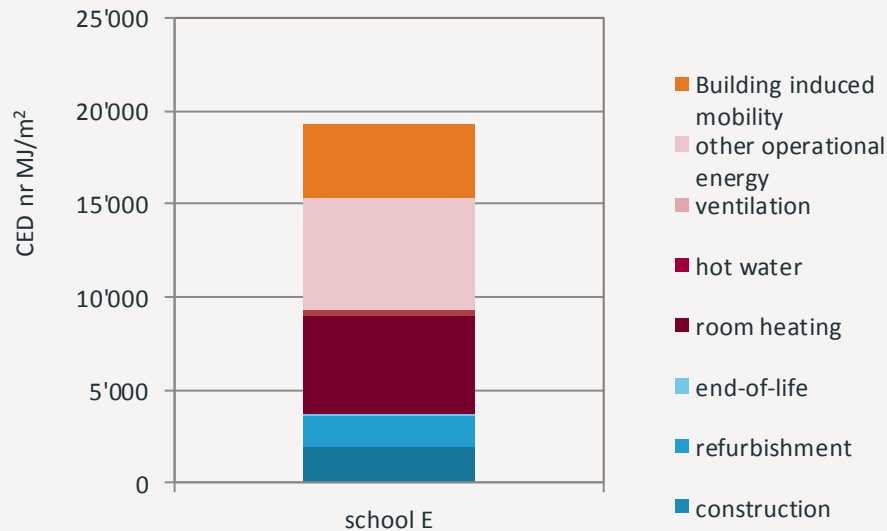


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school E.

Total construction (construction, renewal and deconstruction): The total construction has a share of 19 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 51 % to this. The main impacts come from the roof, windows, flooring as well as the infrastructure. The renewal phase contributes to 46 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 2 %.

The Swiss reference value regarding primary energy demand (non renew.) is slightly exceeded.

Operation: With 61 % the total operation has the main impact on the primary energy demand. The room heating has a share of 46 % and the residual electricity demand causes 52 % of the operational impacts. The Swiss reference level for refurbished school buildings is not exceeded.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 20 % and lies 17 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished school building meets the Swiss target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school E per m² energy reference area and 60 years lifespan.

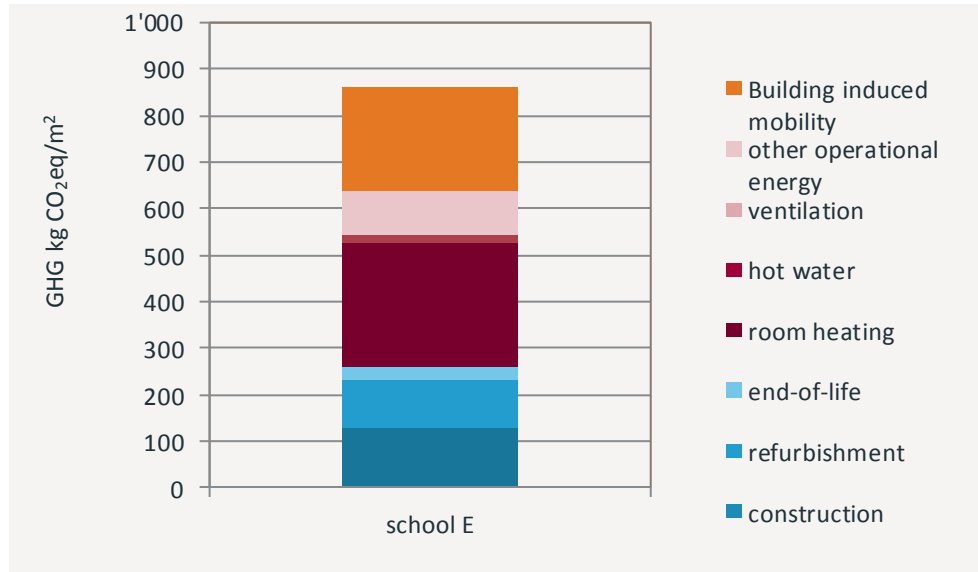


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school E.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 30 %. Within the construction phase the materialization is dominating the GWP (49 %). The main impacts come from the roof, the windows as well as the infrastructure. The renewal per year contributes about 40 % to the GWP of the total construction. The deconstruction has a share of 11 %.

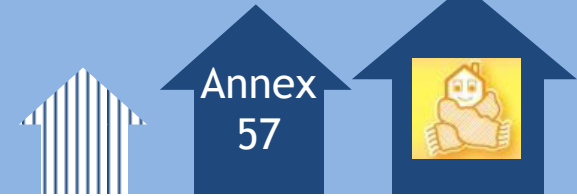
Operation: In the building's use phase the GWP (44 %) is mainly influenced by the room heating and the electricity demand (71 % resp. 25%). The property school E is heated with wood pellets. The reference level is exceeded by 26 %.

Induced mobility: The global warming potential of the induced mobility has a share of 26 % and it lies 12 % below the reference value.

Conclusion: The impacts of the use stage on the GWP are highest. The room heating has the main impact. Overall the GWP of the building school E exceeds the target level for refurbished school buildings slightly.

MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School E, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	14'058
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 8'033 m ²
Surface/Volume ratio (m ⁻¹)	n/a
Construction method	Massive construction (stone/concrete/brick construction)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic window ventilation
Heating and cooling system	Heating:wood pellet, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 0 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 239 MJ/m ² a (per energy reference area) Hot water 10 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie standard
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

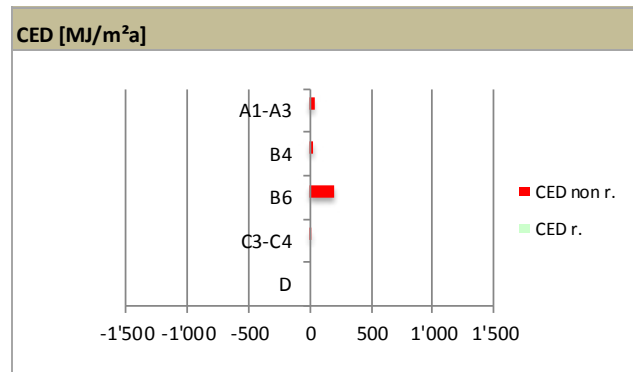
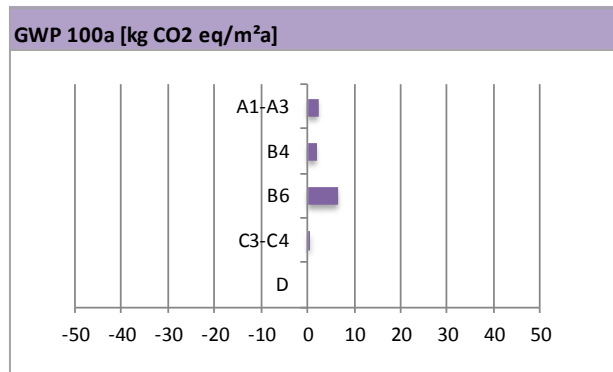
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials uses for the renewal were considered. Building pit Backfill Foundation plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH5	School E	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	2.11E+00	1.74E+00	6.30E+00	4.81E-01	0.00E+00
CED non r.	MJ/m ² a	3.15E+01	2.86E+01	1.94E+02	1.54E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



Inventories and results per building

2.6 School F



The total environmental impacts of buildings

KEY OBSERVATIONS

The school building F (new school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school F has a total env. impact of 36'000 eco-points/m²a, primary energy demand (non renew.) of 440 MJ/m²a and a GWP of 24 kg CO₂/m²a. The building does not meet the target values for new schools regarding global warming potential nor for primary energy demand (non renew). For the indicator CED the construction stage and the use stage cause the same level of impact, whereas the construction stage clearly dominates the greenhouse gas emissions and the total environmental impacts.

The roof, ceilings, the fundament, pillars, flooring and the infrastructure cause the main impact within the construction stage (for CED and GWP) whereas the electricity demand (CED) and room heating (GWP, from district heat) causes the largest impact in the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school F is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: School

Size: Gross floor area 9'582 m², energy reference area 9'279 m²

Location: Zurich, Switzerland

Building year: Completed 2009



ABBREVIATIONS

CED cumulative energy demand
GHG greenhouse gases
GWP global warming potential
LCA life cycle assessment
nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

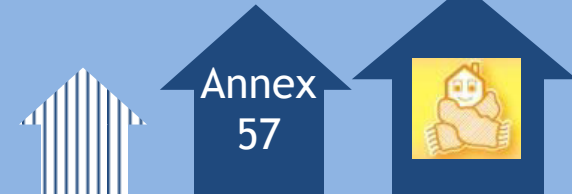
- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
- Frischknecht R., Steiner R. and Jungbluth N. (2008) Methode der ökologischen Knappheit - Ökofaktoren 2006. Umwelt-Wissen Nr. 0906. Bundesamt für Umwelt (BAFU), Bern, retrieved from: www.bafu.admin.ch/publikationen/publikation/01031/index.html?lang=de.
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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School F

The building F is a school property built in 2009. It is a tower-like steel construction glazed all around. The rooms are lying upon another. 22 class rooms, a double sports hall, media center, library, auditorium, canteen, kindergarten and studios build the second largest school building in Zurich.

The heat and hot water demand are covered by district heat. The heat is distributed by radiators. The building is equipped with an automatic ventilation.

CHARACTERISTIC FACTORS OPERATION

Floor area	9'582
Energy reference area	9'279 m ²
Energy demand room heating	53 MJ/m ² a
Energy demand hot water	20 MJm ² a
Energy demand electrical power	45 MJ/m ² a
- Energy demand ventilation:	7 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	Agglomeration
Public transport	grade B
Parking spots	0.74 parking spots per employee
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)



DETAILED RESULTS OF SCHOOL F, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school F in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	9.64	578	578	-	-	0.1	7.0	7.0	-	-	0.01	0.46	0.46	-	-
	backfill	1	86	86	-	-	0.0	1.0	1.0	-	-	0.00	0.07	0.07	-	-
	fundament	2'604	156'238	138'266	-	17'972	16.9	1'013.2	890.1	-	123.1	1.85	111.14	101.75	-	9.38
	ceiling	4'942	296'543	270'355	-	26'189	39.2	2'351.3	2'152.2	-	199.1	3.57	213.98	203.45	-	10.53
	roof	2'290	137'382	86'291	40'425	10'667	23.7	1'420.6	861.7	549.1	9.8	1.65	99.02	51.40	32.37	15.25
	pillars	4'293	257'597	257'597	-	-	24.5	1'470.9	1'470.9	-	-	1.44	86.36	86.36	-	-
	outer walls basement	493	29'555	25'402	-	4'153	3.7	224.8	207.7	-	17.0	0.41	24.78	20.89	-	3.88
	outer walls upper floors	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	windows	1'335	80'127	38'681	38'681	2'764	18.5	1'109.5	546.7	546.7	16.1	1.39	83.12	40.16	40.16	2.80
	inner walls raw	715	42'874	40'068	-	2'805	7.2	432.2	408.1	-	24.1	0.62	37.36	36.14	-	1.22
	separation walls/inner doors	129	7'722	3'651	3'651	421	1.2	69.8	33.8	33.8	2.2	0.07	4.29	2.09	2.09	0.12
	flooring	1'914	114'830	44'012	44'012	26'807	22.3	1'336.0	631.5	631.5	73.1	1.96	117.73	46.16	46.16	25.41
	wall cover	35	2'120	1'046	1'046	29	0.6	36.1	18.0	18.0	0.2	0.03	1.68	0.84	0.84	0.01
	ceiling cover	136	8'143	3'655	3'655	832	1.7	103.4	50.5	50.5	2.3	0.09	5.59	2.74	2.74	0.11
	infrastructure	4'337	260'192	124'333	127'633	8'226	25.8	1'545.8	753.5	779.8	12.5	1.60	96.21	41.45	43.06	11.70
	sum building	23'233	1'393'988	1'034'021	259'103	100'865	185.4	11'121.6	8'032.7	2'609.4	479.5	14.70	881.81	633.98	167.41	80.41
Operation	room heating	1'379	82'770				44.2	2'654.4				2.48	148.62			
	hot water	761	45'650				24.4	1'464.0				1.37	81.97			
	ventilation	876	52'569				18.4	1'106.6				0.29	17.31			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	7'773	466'355				187.2	11'231.9				5.70	341.89			
Building induced mobility	sum mobility	4'941	296'439				70.0	4'200.4				3.90	233.85			
sum total	construction, operation und building induced mobility	35'946					442.6					24.29				
target							350.0					14.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school F per m² energy reference area and 60 years lifespan.

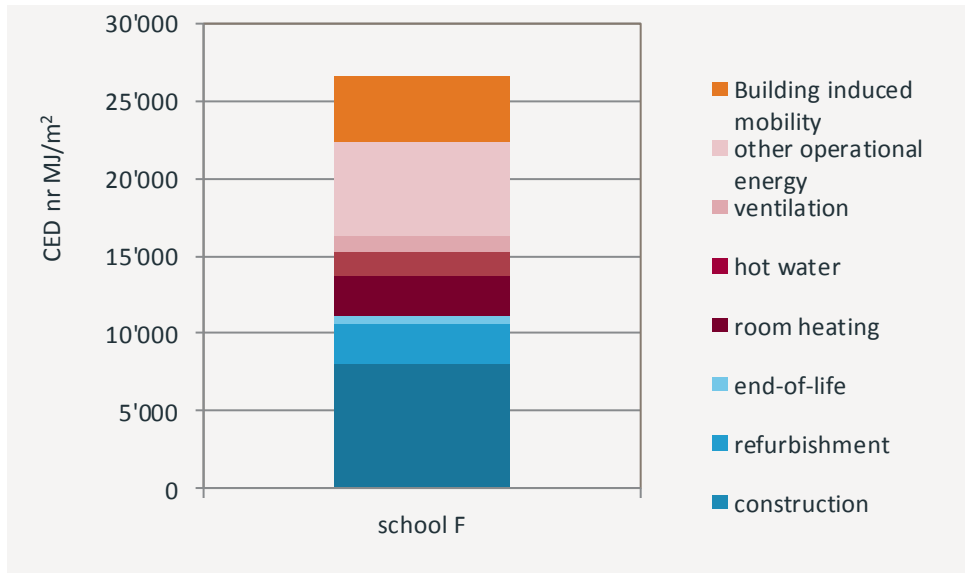


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school F.

Total construction (construction, renewal and deconstruction): The total construction as well as the operation each have a share of 42 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 72 % to this. The main impacts come from the ceiling, the roof, pillars as well as the infrastructure. The renewal phase contributes to 23 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 4 %. The Swiss reference value regarding primary energy demand (non renew.) is exceeded by 70 %.

Operation: The operational energy demand is dominating the environmental impacts of the use stage (53 %), followed by the room heating (24 %). The Swiss reference level for new school buildings is slightly exceeded.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 16 % and lies 17 % above the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the newly constructed school building does not meet the Swiss target value. Similar impact are caused by the construction and the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school F per m² energy reference area and 60 years lifespan.

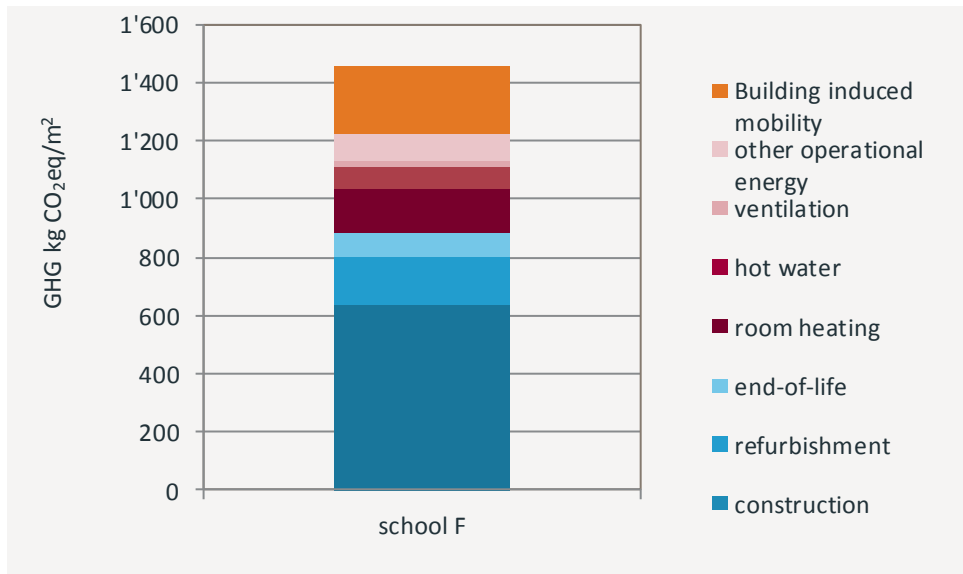


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school F.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 60 %. Within the construction phase the materialization is dominating the GWP (72 %). The main impacts come from the fundament, the ceilings and the roof. The renewal per year contributes about 19 % to the GWP of the total construction. The deconstruction has a share of 9 %.

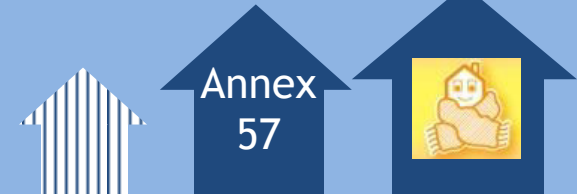
Operation: In the building's use phase the GWP (23 %) is mainly influenced by the room heating and the electricity demand (43% resp. 27%). The property school F is heated with district heat. The reference level is exceeded by 128 %.

Induced mobility: The global warming potential of the induced mobility has a share of 16 % and it lies 30 % above the reference value.

Conclusion: The impacts of the construction stage on the GWP are dominating the results. The construction has the main impact. Overall the GWP of the building school F exceeds the target level for new school buildings by 70 %.

MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School F, new construction
Energy-standard	net-positive
Gross floor area/ Net floor area	9'582 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 9'279 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (glass/steel)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating:district heat, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 7 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 53 MJ/m ² a (per energy reference area) Hot water 20 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	-
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

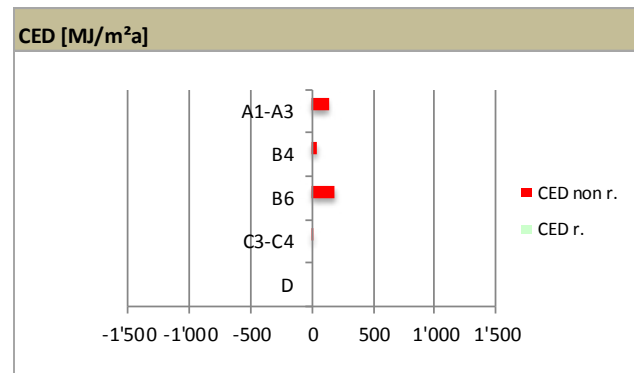
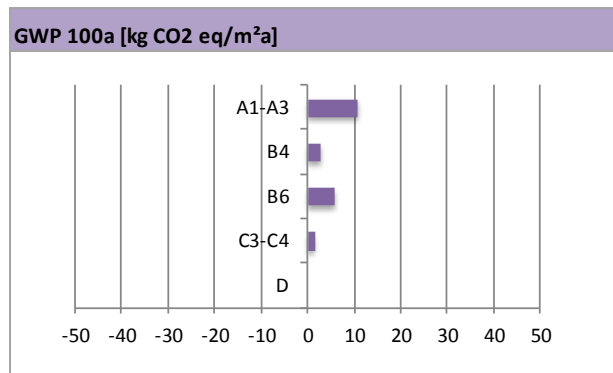
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Building pit Backfill Foundation plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH6	School F	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	1.06E+01	2.79E+00	5.70E+00	1.34E+00	0.00E+00
CED non r.	MJ/m ² a	1.34E+02	4.35E+01	1.87E+02	7.99E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



Inventories and results per building

2.7 School G



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The school building (newly constructed school building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The school G has a total env. impact of 24'000 eco-points/m²a, primary energy demand (non renew.) of 350 MJ/m²a and a GWP of 12 kg CO₂/m²a. The building meets the target values for newly constructed schools regarding global warming potential and for primary energy demand (non renew). The indicator CED is dominated by the use stage, while the construction stage is most important for the greenhouse gas emissions.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This school G is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: School

Size: gross floor area not known, 504 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2013



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



School G

30 school pavilions of this type are used in the city of Zurich actually. These help to mitigate school shortage until proper school buildings are built. Pavilions like school G fulfill the criteria of Minergie. Windows are automatized for ventilation.

The heat and hot water demand is covered by an electric heat pump equipped with a borehole heat exchanger. The heat distribution works with radiators.

CHARACTERISTIC FACTORS OPERATION

Floor area	n/a
Energy reference area	504 m ²
Energy demand room heating	53 MJ/m ² a
Energy demand hot water	20 MJm ² a
Energy demand electrical power	45 MJ/m ² a
- Energy demand ventilation:	7 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	-
Public transport	-
Parking spots	0.74 parking spots per employee (Swiss average)
Public transport subscriptions	0.22 permanent public transport subscriptions (Swiss average)
Bicycle parking	1 (available)



DETAILED RESULTS OF SCHOOL G, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the school G in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0.51	30	30	-	-	0.0	0.4	0.4	-	-	0.00	0.02	0.02	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	254	15'261	13'209	-	2'052	1.6	94.8	79.2	-	15.6	0.19	11.16	10.32	-	0.84
	ceiling	1'931	115'883	56'068	29'719	30'096	20.8	1'245.0	801.7	430.8	12.5	1.04	62.56	34.51	18.51	9.55
	roof	1'112	66'714	34'196	21'830	10'688	13.9	836.0	513.9	316.8	5.3	0.74	44.20	24.56	16.69	2.95
	pillars	445	26'721	18'053	-	8'667	3.4	204.2	201.1	-	3.0	0.21	12.67	10.51	-	2.16
	outer walls basement	697	41'842	17'405	17'405	7'032	7.8	466.0	230.3	230.3	5.4	0.51	30.86	14.26	14.26	2.33
	outer walls upper floors	1'481	88'863	41'760	26'177	20'927	16.9	1'016.5	633.0	373.8	9.7	0.78	47.06	25.91	15.88	5.28
	windows	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	inner walls raw	371	22'264	7'215	7'215	7'834	3.4	201.5	99.4	99.4	2.6	0.17	10.42	4.49	4.49	1.43
	separation walls/inner doors	479	28'731	9'012	9'950	9'769	4.4	263.2	121.2	137.9	4.0	0.27	15.94	5.92	6.62	3.41
	flooring	355	21'289	8'339	8'339	4'612	3.0	180.8	89.7	89.7	1.3	0.19	11.51	5.39	5.39	0.73
	wall cover	528	31'673	9'577	11'676	10'421	4.3	259.7	109.6	146.9	3.2	0.30	17.76	5.62	7.17	4.97
	ceiling cover	514	30'856	8'890	11'596	10'370	4.3	257.2	103.0	151.2	3.0	0.30	18.14	5.24	7.25	5.65
	infrastructure	3'211	192'668	90'569	93'870	8'229	21.7	1'301.7	631.4	657.8	12.5	1.36	81.66	34.17	35.78	11.70
	sum building	11'380	682'796	314'323	237'776	130'697	105.4	6'326.8	3'613.9	2'634.7	78.2	6.07	363.96	180.93	132.04	51.00
Operation	room heating	2'051	123'049				42.3	2'538.8				0.66	39.72			
	hot water	1'400	84'030				29.3	1'756.4				0.46	27.48			
	ventilation	876	52'569				18.4	1'106.6				0.29	17.31			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	9'084	545'014				190.1	11'408.7				2.97	178.50			
Building induced mobility	sum mobility		3'742	224'544			55.5	3'329.1				3.03	181.54			
sum total	construction, operation und building induced mobility		24'206				351.1					12.07				
target							350.0					14.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the building school G per m² energy reference area and 60 years lifespan.

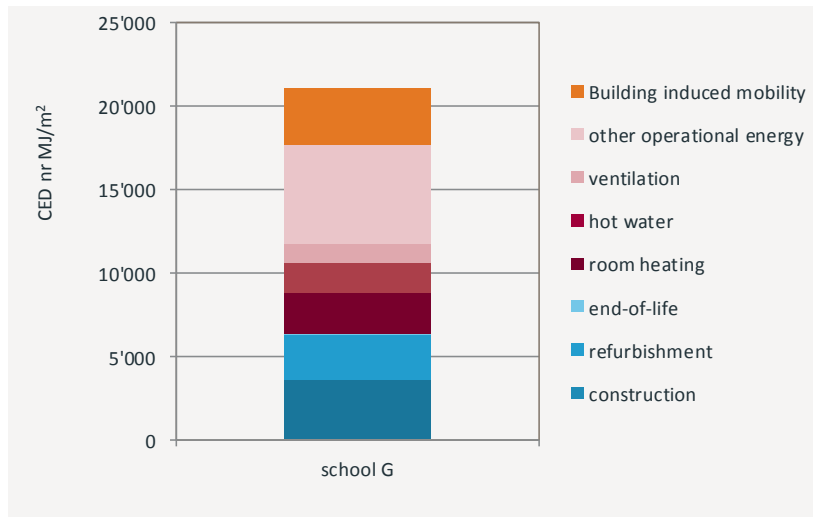


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the building school G.

Total construction (construction, renewal and deconstruction): The total construction has a share of 30 % to the overall primary energy demand (non renew.).

The construction phase itself contributes to 57 % to this. The main impacts come from the fundement, the outer walls, the roof as well as the infrastructure. The renewal phase contributes to 42 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 1 %.

The Swiss reference value regarding primary energy demand (non renew.) is met.

Operation: With 54 % the total operation has the main impact on the primary energy demand. The room heating has a share of 22 % and the residual electricity demand causes 53 % of the impacts. The Swiss reference level for new school buildings is slightly exceeded.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 16 % to the CED and lies 8 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the new school meets the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the building school G per m² energy reference area and 60 years lifespan.

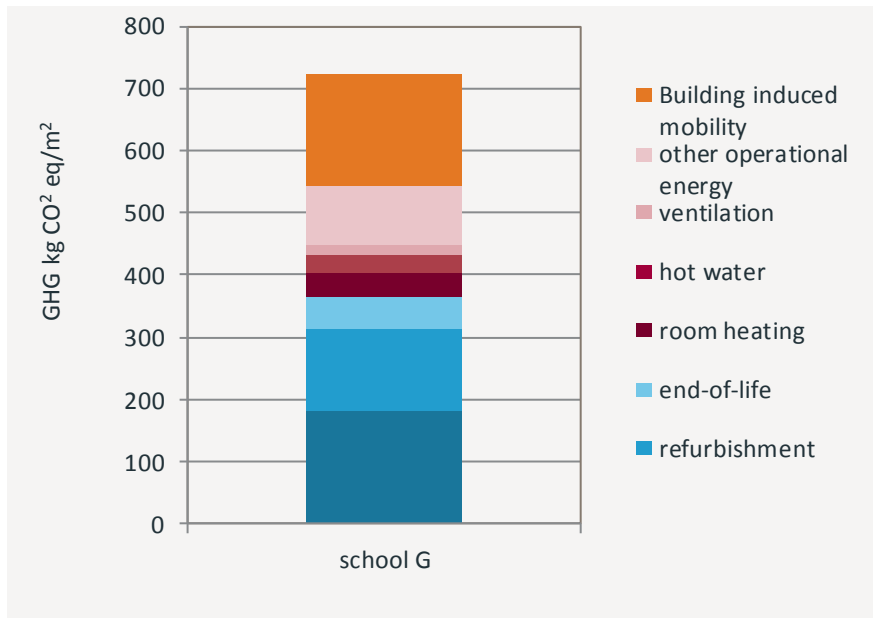


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the building school G.

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 50 %. Within the construction phase the materialization is dominating the GWP (50 %). The main impacts come from the fundement, walls, the roof as well as the infrastructure. The renewal per year contributes about 36 % to the GWP of the total construction. The deconstruction has a share of 14 %. The reference level regarding GWP for new schools is not exceeded.

Operation: In the building's use phase (25 %) the GWP is mainly influenced by the additional electricity demand (53 %). The property school G is heated with a borehole heat exchanger. The reference level is exceeded by 20 %.

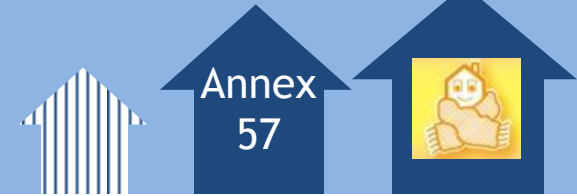
Induced mobility: The global warming potential of the induced mobility has a share of 25 % on the total GHG emissions and meets the reference value.

Conclusion: The impacts of the construction stage on the GWP are most dominant. The construction itself has the main impact. Overall the GWP of the building school G holds the target level for new school buildings.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	school building – School G, new construction
Energy-standard	net-positive
Gross floor area/ Net floor area	n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 504 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Modular concept (mixed construction: wood, glass)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 7 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 53 MJ/m ² a (per energy reference area) Hot water 20 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

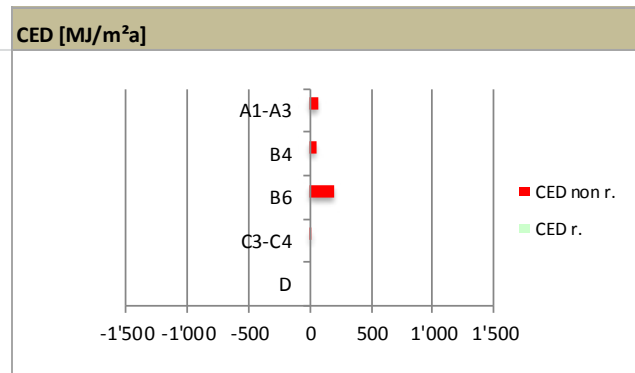
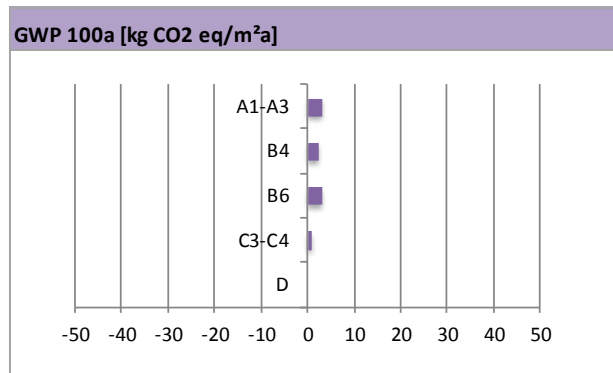
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Building pit Backfill Foundation plate Ceilings Roof External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH7	School G	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	3.02E+00	2.20E+00	2.97E+00	8.50E-01	0.00E+00
CED non r.	MJ/m ² a	6.02E+01	4.39E+01	1.90E+02	1.30E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



2.8 Residential building A



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The residential building A (refurbished building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The residential building A has a total env. impact of 27'000 eco-points/m²a, primary energy demand (non renew.) of 490 MJ/m²a and a GWP of 17 kg CO₂/m²a. The building does not meet the target values for refurbished residential buildings regarding global warming potential nor for primary energy demand (non renew). All indicators are dominated by the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This residential building A is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: Residential home

Size: 5'259 m² gross floor area, 4'097 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed 2013



ABBREVIATIONS

CED cumulative energy demand
GHG greenhouse gases
GWP global warming potential
LCA life cycle assessment
nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

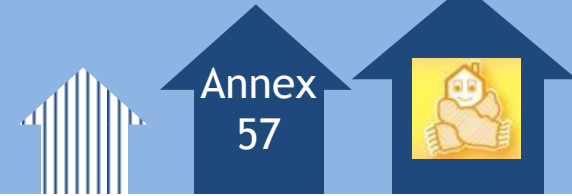
- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
- Frischknecht R., Steiner R. and Jungbluth N. (2008) Methode der ökologischen Knappheit - Ökofaktoren 2006. Umwelt-Wissen Nr. 0906. Bundesamt für Umwelt (BAFU), Bern, retrieved from: www.bafu.admin.ch/publikationen/publikation/01031/index.html?lang=de.
- Frischknecht R., Jungbluth N., Althaus H.-J., Bauer C., Doka G., Dones R., Hellweg S., Hischer R., Humbert S., Margni M. and Nemecek T. (2007) Implementation of Life Cycle Impact Assessment Methods. ecoinvent report No. 3, v2.0. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
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- SIA D 0236 (2011), SIA-Effizienzpfad Energie – Ergänzungen und Fallbeispiele zum Merkblatt SIA 2040

Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Residential building A

The residential property A consists of building with five upper floors, a ground floor and a basement. It's 42 years old. The building is brick-built with a flat roof. Within the refurbishment the façade, the technical installations and the interior are renewed. The energy demand of the use phase should be reduced by 75 % by the renewal of the façade.

The heat and hot water demand is covered by an electric heat pump equipped with a borehole heat exchanger. A gas-fueled boiler covers the peak demand.

CHARACTERISTIC FACTORS OPERATION

Floor area	5'259 m ²
Energy reference area	4'097 m ²
Energy demand room heating	135 MJ/m ² a
- Heat pump	67 %
- Gas	34 %
Energy demand hot water	50 MJm ² a
- Heat pump	80 %
- Gas	20 %
Energy demand electrical power	49 MJ/m ² a
- Energy demand ventilation:	10 MJ/m ² a
- Energy demand residual operation:	39 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Parking spots	0.33 parking spots per household
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)
Private cars per person	0.36 cars per person



DETAILED RESULTS OF THE RESIDENTIAL BUILDING A, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the residential building A in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0	0	0	-	-	0.0	0.0	0.0	-	-	0.00	0.00	0.00	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	ceiling	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	roof	312	18'747	6'297	3'149	9'301	5.0	300.5	192.5	96.3	11.7	0.35	20.95	5.52	2.76	12.68
	pillars	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls basement	42	2'526	2'190	-	336	0.3	15.7	13.2	-	2.5	0.03	1.83	1.69	-	0.14
	outer walls upper floors	554	33'267	12'096	10'469	10'702	9.6	577.6	289.2	279.3	9.1	0.74	44.45	15.00	13.82	15.63
	windows	1'178	70'692	31'631	31'631	7'430	14.5	869.5	431.1	431.1	7.2	0.99	59.52	28.45	28.45	2.62
	inner walls raw	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	separation walls/inner doors	144	8'625	3'187	3'187	2'251	1.3	79.9	39.1	39.1	1.7	0.08	4.86	2.07	2.07	0.71
	flooring	198	11'871	5'793	5'793	286	1.2	72.4	35.1	35.1	2.2	0.08	4.84	2.36	2.36	0.11
	wall cover	257	15'394	6'871	6'871	1'653	1.9	111.1	54.6	54.6	2.0	0.13	7.96	3.07	3.07	1.83
	ceiling cover	187	11'204	4'698	4'698	1'808	2.2	131.6	63.9	63.9	3.7	0.13	7.58	2.95	2.95	1.68
	infrastructure	3'153	189'205	89'408	92'708	7'089	18.7	1'121.8	541.3	567.6	13.0	1.18	70.60	29.53	31.14	9.93
	sum building	6'025.5	361'532	162'172	158'505	40'856	54.7	3'280.1	1'660.1	1'567.1	53.0	3.71	222.58	90.64	86.62	45.32
Operation	room heating	5'277	316'619				134.2	8'053.5				4.79	287.69			
	hot water	3'337	200'219				77.3	4'635.2				2.02	121.29			
	ventilation	1'252	75'098				26.3	1'580.8				0.41	24.73			
	other operational energy	4'881	292'877				102.7	6'165.0				1.61	96.45			
	sum operation	14'747	884'814				340.6	20'434.5				8.84	530.17			
Building induced mobility	sum mobility	6'096	365'750				93.0	5'580.0				4.83	290.00			
sum total	construction, operation und building induced mobility	26'868					488.2					17.38				
target							440.0					15.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building A per m² energy reference area and 60 years lifespan.

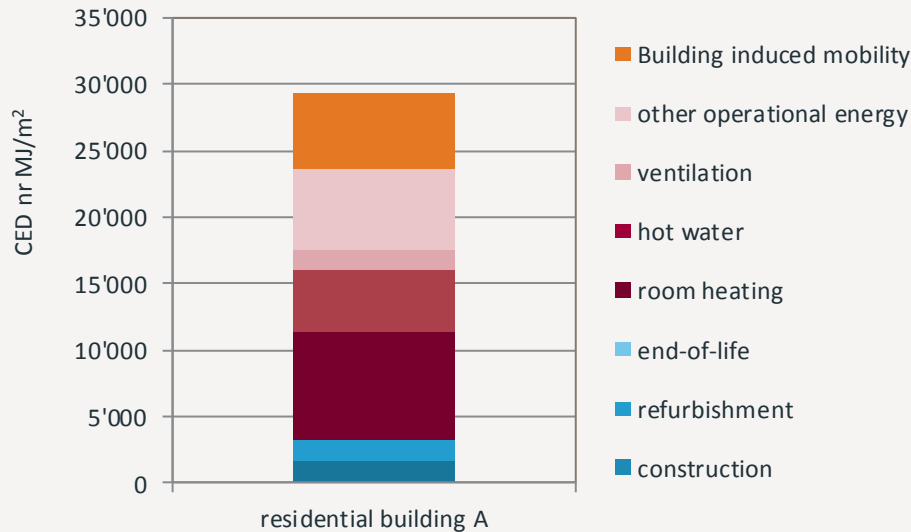


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building A .

Total construction (construction, renewal and deconstruction): The total construction has a share of 11 % to the overall primary energy demand (non renew.).

The construction itself contributes to 51 % to this. The main impacts come from the windows, the walls as well as from the infrastructure. The renewal phase contributes to 48 % to the primary energy demand of the total construction. In comparison the deconstruction has only an impact of 1 %.

The Swiss reference value regarding primary energy demand (non renew.) is met.

Operation: With 70 % the total operation has the main impact on the primary energy demand. The room heating has a share of 39 % and the residual electricity demand causes 30 % of the impacts. The Swiss reference level for refurbished residential buildings is exceeded by 36 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 19 % to the CED and lies 28 % lower than the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished residential property exceeds the target value by 11 %. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the residential building A per m² energy reference area and 60 years lifespan.

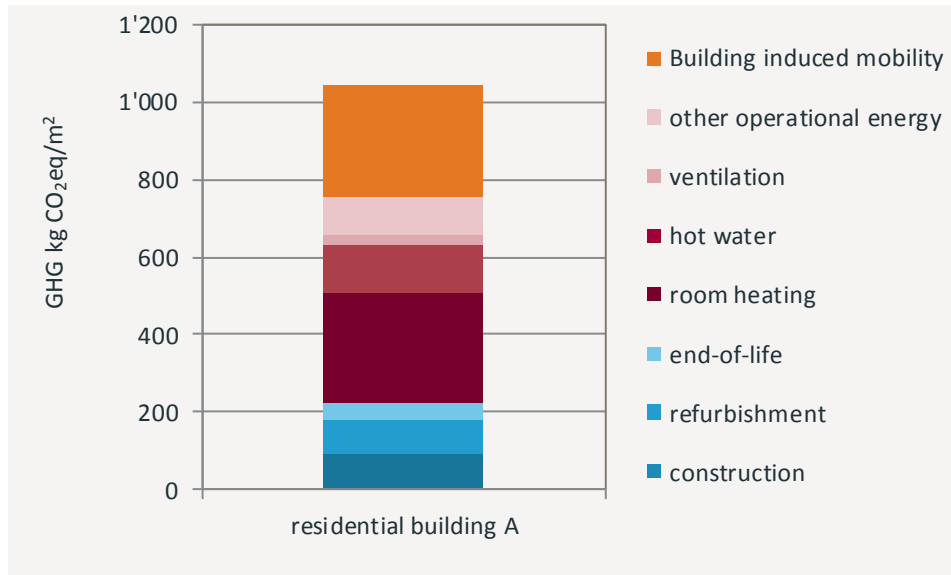


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the residential building A .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 21 %. Within the construction phase the materialization is dominating the GWP (41 %). The main impacts come from the walls, windows as well as from the infrastructure. The renewal per year contributes about 39 % to the GWP of the total construction. The deconstruction has a share of 20 %. The reference level regarding GWP for refurbished residential buildings is not exceeded.

Operation: In the building's use phase (51 %) the GWP is mainly influenced by the room heating (54 %), followed by the hot water provision (23 %). The residential building A is heated with a borehole heat exchanger. Peak demands are covered by a gas-fueled boiler. The reference level is exceeded by 77 %.

Induced mobility: The global warming potential of the induced mobility has a share of 28 % on the total GHG emissions and meets the reference value.

Conclusion: The impacts of the use stage on the GWP are most dominant. The room heating has the main impact. Overall the GWP of the residential building A exceeds the target level for refurbished residential buildings.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Residential A, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	5'259 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 4'097 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, a gas-fueled boiler covers the peak demand. heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 10 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 39 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 135 MJ/m ² a (per energy reference area) - Heat pump 67 % - Gas 34 % Hot water 50 MJ/m ² a (per energy reference area) - Heat pump 80 % - Gas 20 %
Final energy demand for cooling	0 MJ/m ² a
Benchmark	-
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Roof External walls (underground and above ground) Windows, doors Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	-
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

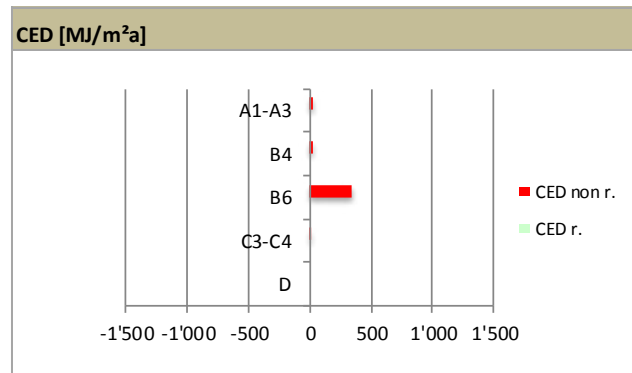
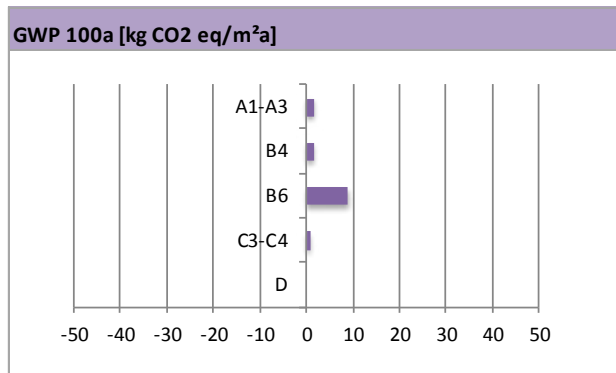
MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment	
CH8	Residential building A

57 Annex 57

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	1.51E+00	1.44E+00	8.84E+00	7.55E-01	0.00E+00
CED non r.	MJ/m ² a	2.77E+01	2.61E+01	3.41E+02	8.83E-01	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



2.9 Residential building B



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The residential building B (refurbished building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The residential building B has a total env. impact of 27'000 eco-points/m²a, primary energy demand (non renew.) of 430 MJ/m²a and a GWP of 14 kg CO₂/m²a. The building meets the target values for refurbished residential buildings regarding global warming potential and for primary energy demand (non renew). The CED is dominated by the use stage whereas the GHG emissions are dominated by the construction stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This residential building B is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: Residential home

Size: gross floor area not known, 2'894 m² energy reference area

Location: Zurich, Switzerland

Building year: Completion in 2016



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

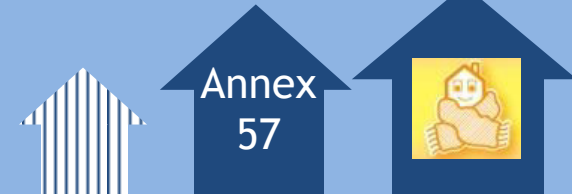
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- Frischknecht R., Steiner R. and Jungbluth N. (2008) Methode der ökologischen Knappheit - Ökofaktoren 2006. Umwelt-Wissen Nr. 0906. Bundesamt für Umwelt (BAFU), Bern, retrieved from: www.bafu.admin.ch/publikationen/publikation/01031/index.html?lang=de.
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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building Bre calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Residential building B

The residential building B was constructed 1970-1972 from the architect Erwin Müller. The property consists of several buildings with total 220 flats. The 40 years old buildings are now renewed, in which small flats are reconstructed to larger, family-friendly apartments. The refurbishment covers the façade, the interior as well as the infrastructure.

The heat and hot water demand is covered by an electric heat pump equipped with a borehole heat exchanger.

CHARACTERISTIC FACTORS OPERATION

Floor area	n/a
Energy reference area	2'894 m ²
Energy demand room heating	77 MJ/m ² a
Energy demand hot water	50 MJm ² a
Energy demand electrical power	49 MJ/m ² a
- Energy demand ventilation:	10 MJ/m ² a
- Energy demand residual operation:	39 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade C
Distance for groceries	0.1 km
Parking spots	0.4 parking spots per household
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)
Private cars per person	0.36 cars per person (average city of Zurich)



DETAILED RESULTS OF THE RESIDENTIAL BUILDING B, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the residential building B in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ²	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	0.34	21	21	-	-	0.0	0.2	0.2	-	-	0.00	0.02	0.02	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	13	793	581	-	212	0.1	5.5	4.3	-	1.2	0.01	0.53	0.39	-	0.14
	ceiling	21	1'232	911	-	321	0.3	16.3	14.2	-	2.1	0.01	0.77	0.54	-	0.22
	roof	365	21'881	11'950	8'562	1'368	2.3	140.7	84.4	54.3	1.9	0.14	8.61	4.25	2.85	1.51
	pillars	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls basement	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls upper floors	441	26'444	12'255	7'275	6'914	6.5	389.6	241.7	140.6	7.3	0.56	33.90	16.02	8.84	9.03
	windows	3'959	237'560	103'985	103'985	29'591	43.1	2'583.3	1'283.4	1'283.4	16.4	2.96	177.35	82.53	82.53	12.28
	inner walls raw	14	819	702	-	118	0.1	6.9	6.0	-	0.9	0.01	0.76	0.71	-	0.05
	separation walls/inner doors	506	30'362	14'402	14'402	1'558	3.6	214.9	106.1	106.1	2.6	0.20	11.88	5.69	5.69	0.51
	flooring	385	23'116	9'329	9'781	4'005	3.0	181.3	85.7	93.5	2.1	0.17	10.41	4.32	4.65	1.45
	wall cover	590	35'382	9'726	17'622	8'034	5.1	306.6	83.4	219.8	3.4	0.39	23.68	4.14	9.88	9.66
	ceiling cover	273	16'406	4'840	7'819	3'746	2.4	142.8	42.3	93.8	6.6	0.17	10.47	2.21	4.37	3.89
	infrastructure	1'674	100'462	46'900	50'200	3'361	11.0	660.9	314.8	341.1	5.0	0.70	41.71	17.68	19.28	4.75
	sum building	8'241	494'478	215'602	219'647	59'229	77.5	4'649.1	2'266.8	2'332.8	49.6	5.33	320.09	138.50	138.11	43.48
Operation	room heating	2'534	152'047				52.0	3'121.0				0.81	48.83			
	hot water	3'157	189'393				65.9	3'951.9				1.03	61.83			
	ventilation	1'252	75'098				26.3	1'580.8				0.41	24.73			
	other operational energy	4'881	292'877				102.7	6'165.0				1.61	96.45			
	sum operation	11'824	709'416				247.0	14'818.7				3.86	231.85			
Building induced	sum mobility	6'618	397'050				101.2	6'070.0				5.17	310.00			
sum total	construction, operation und building induced mobility	26'682					425.6					14.37				
target							440.0					15.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building B per m² energy reference area and 60 years lifespan.

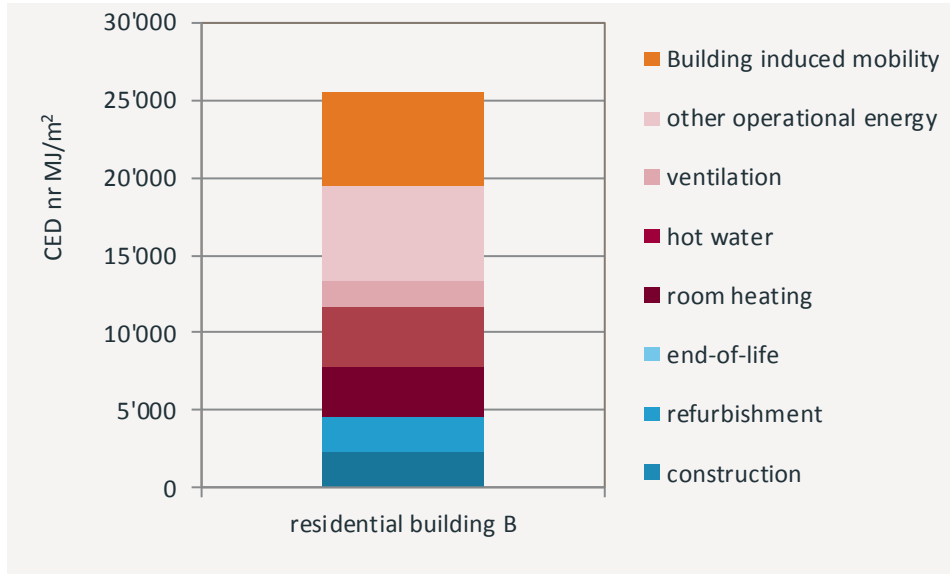


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building B .

Total construction (construction, renewal and deconstruction): The total construction has a share of 18 % to the overall primary energy demand (non renew.).

The construction itself contributes to 49 % to this. The main impacts come from the windows (aluminium-wood frames), the walls, wall covers as well as from the infrastructure. The renewal phase contributes to 50 % to the primary energy demand of the total construction and is influenced by the same materials. In comparison the deconstruction has only an impact of 1 %. The Swiss reference value regarding primary energy demand (non renew.) is met.

Operation: With 58 % the total operation has the main impact on the primary energy demand. The largest impact is caused by the electricity demand (42 %), followed by the hot water provision (27 %). The Swiss reference level for refurbished residential buildings is exceeded by 23 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 24 % to the CED and holds the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished residential property meets the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the residential building B per m² energy reference area and 60 years lifespan.

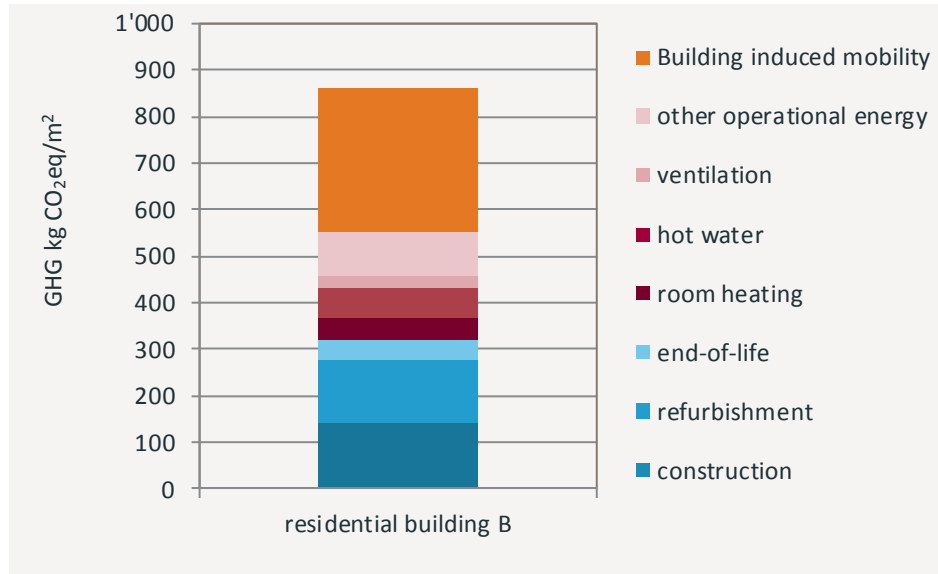


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the residential building B .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 37 %. Within the construction phase the materialization is dominating the GWP (43 %). The main impacts come from the walls, windows as well as from the infrastructure. Especially the windows with aluminium-wood frames dominate the GHG emissions. The renewal per year contributes about 43 % to the GWP of the total construction. The aluminium-wood framed windows have the main impact as well. The deconstruction has a share of 14 %. The reference level regarding GWP for refurbished residential buildings is not exceeded.

Operation: In the building's use phase (27 %) the GWP is mainly influenced by electricity demand (42 %), followed by the hot water provision (27 %). The residential building B is heated with a borehole heat exchanger. The reference level is exceeded by 55 %.

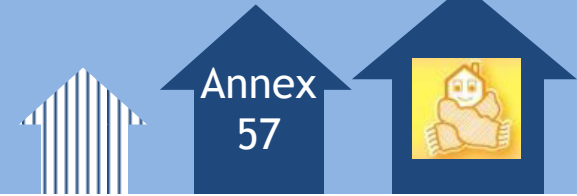
Induced mobility: The global warming potential of the induced mobility has a share of 36 % on the total GHG emissions and meets the reference value.

Conclusion: The impacts of the construction stage on the GWP are most dominant. The construction and the refurbishment have similar impacts and dominate the results. Overall the GWP of the residential building B meets the target level for refurbished residential buildings.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Residential B, new construction
Energy-standard	net-positive
Gross floor area/ Net floor area	n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 2'894 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, heat distribution with radiators Cooling: n/a
Final energy demand electricity	Ventilation 10 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 39 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 77 MJ/m ² a (per energy reference area) Hot water 50 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	-
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

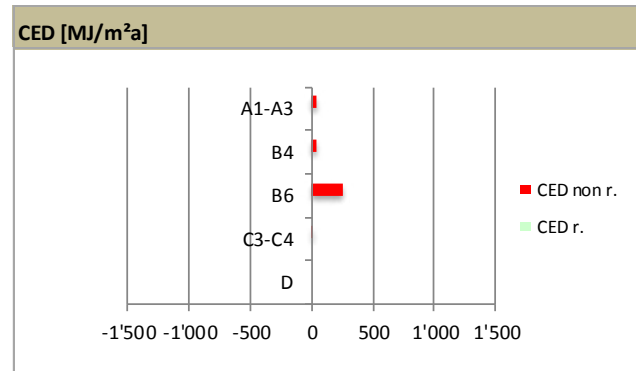
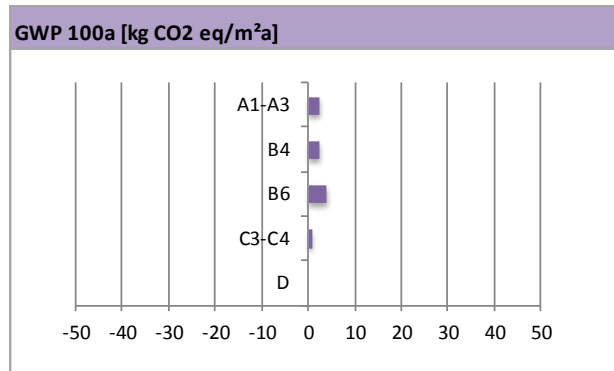
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Building pit Backfill Foundation plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH9	Residential building B	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	2.31E+00	2.30E+00	3.86E+00	7.25E-01	0.00E+00
CED non r.	MJ/m ² a	3.78E+01	3.89E+01	2.47E+02	8.26E-01	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



2.10 Residential building E



The total environmental impacts of buildings

KEY OBSERVATIONS

The residential building E (new building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The residential building E has a total env. impact of 31'000 eco-points/m²a, primary energy demand (non renew.) of 440 MJ/m²a and a GWP of 18 kg CO₂/m²a. The building meets the target values for new residential buildings regarding CED nr but not for the greenhouse gas emissions. The CED is dominated by the use stage whereas the GHG emissions are dominated by the construction stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This residential building E is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: Residential home

Size: 2'478 m² gross floor area, 2'156 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed in 2014



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

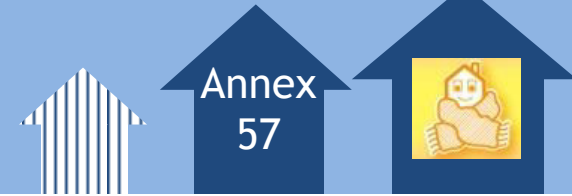
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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Residential building E

The new residential property is going to be built in Zurich-Altstetten providing 104 apartments. 7 new buildings replace the four old ones, which are deconstructed. The heat and hot water demand is covered by an electric heat pump equipped with a borehole heat exchanger.

CHARACTERISTIC FACTORS OPERATION

Floor area	2'487 m ²
Energy reference area	2'156 m ²
Energy demand room heating	57 MJ/m ² a
Energy demand hot water	50 MJm ² a
Energy demand electrical power	45 MJ/m ² a
- Energy demand ventilation:	6 MJ/m ² a
- Energy demand residual operation:	39 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade C
Distance for groceries	0.4 km
Parking spots	0.5 parking spots per household
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)
Private cars per person	0.36 cars per person (average city of Zurich)

DETAILED RESULTS OF THE RESIDENTIAL BUILDING E, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the residential building E in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ²	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	12.59	756	756	-	-	0.2	9.1	9.1	-	-	0.01	0.61	0.61	-	-
	backfill	4	257	257	-	-	0.1	3.1	3.1	-	-	0.00	0.21	0.21	-	-
	fundament	744	44'639	38'141	-	6'498	4.7	279.4	230.4	-	49.0	0.42	25.08	22.47	-	2.61
	ceiling	1'855	111'301	97'739	-	13'562	11.4	684.9	585.7	-	99.2	1.27	76.07	70.62	-	5.45
	roof	1'028	61'699	46'930	2'941	11'828	8.7	521.2	399.2	77.0	45.0	0.76	45.70	31.97	2.41	11.31
	pillars	34	2'032	1'878	-	155	0.2	12.0	10.9	-	1.2	0.02	1.11	1.04	-	0.06
	outer walls basement	258	15'454	12'205	1'077	2'173	1.8	110.2	82.6	16.0	11.5	0.22	13.28	9.87	1.82	1.59
	outer walls upper floors	513	30'777	24'299	-	6'479	7.4	443.3	422.7	-	20.6	0.62	37.31	30.35	-	6.95
	windows	1'545	92'695	40'429	40'429	11'838	18.6	1'114.8	552.8	552.8	9.2	1.26	75.86	35.41	35.41	5.05
	inner walls raw	701	42'062	37'184	-	4'878	4.6	277.2	244.0	-	33.2	0.47	28.21	26.16	-	2.05
	separation walls/inner doors	1'088	65'292	28'861	28'861	7'569	10.2	612.6	299.5	299.5	13.5	0.64	38.42	17.95	17.95	2.53
	flooring	941	56'435	21'962	21'962	12'510	9.9	593.5	277.7	277.7	38.1	0.89	53.24	22.44	22.44	8.36
	wall cover	378	22'666	10'094	10'094	2'478	3.1	185.2	88.8	88.8	7.6	0.24	14.49	6.29	6.29	1.91
	ceiling cover	109	6'570	2'553	2'553	1'464	1.1	68.5	32.7	32.7	3.0	0.08	4.51	1.58	1.58	1.34
	infrastructure	4'088	245'285	110'898	127'298	7'089	33.2	1'994.0	874.5	1'106.6	13.0	2.03	121.99	49.52	62.54	9.93
	sum building	13'299	797'921	474'185	235'215	88'521	115.2	6'909.0	4'113.8	2'451.1	344.1	8.93	536.07	326.49	150.44	59.14
Operation	room heating	1'876	112'554				38.5	2'310.3				0.60	36.15			
	hot water	3'157	189'393				65.9	3'951.9				1.03	61.83			
	ventilation	751	45'058				15.8	948.5				0.25	14.84			
	other operational energy	4'881	292'877				102.7	6'165.0				1.61	96.45			
	sum operation	10'665	639'883				222.9	13'375.7				3.49	209.27			
Building induced	sum mobility	6'759	405'550				103.2	6'190.0				5.33	320.00			
sum total	construction, operation und building induced mobility	30'723					441.2					17.76				
target							440.0					16.50				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building E per m² energy reference area and 60 years lifespan.

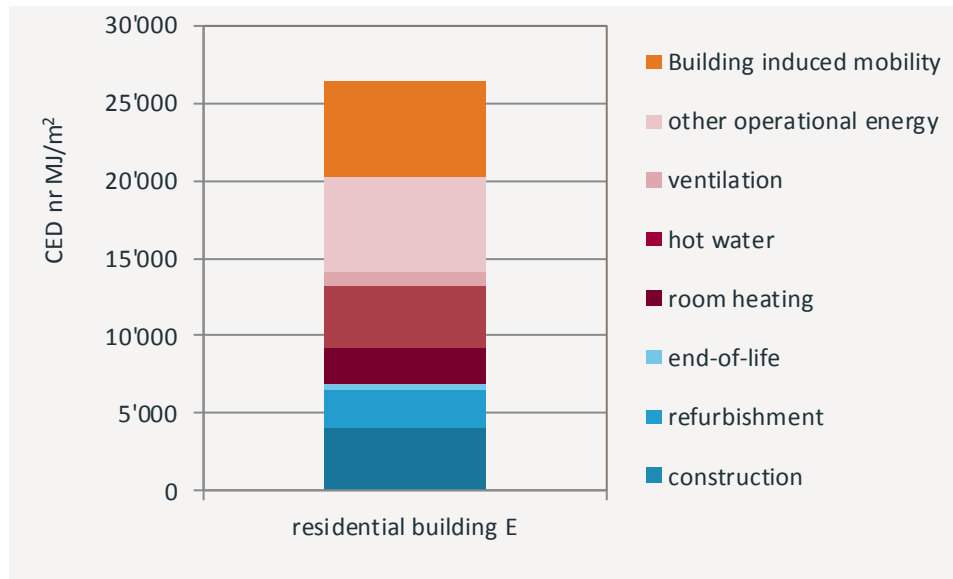


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the residential building E .

Total construction (construction, renewal and deconstruction): The total construction has a share of 26 % to the overall primary energy demand (non renew.).

The construction itself contributes to 60 % to this. The main impacts come from the ceiling, the windows, the walls, the flooring as well as from the infrastructure. The renewal phase contributes to 35 % to the primary energy demand of the total construction. In comparison the deconstruction has an impact of 5 %. The Swiss reference value regarding primary energy demand (non renew.) is slightly exceeded.

Operation: With 51 % the total operation has the main impact on the primary energy demand. The largest impact is caused by the electricity demand (46 %), followed by the hot water provision (30 %). The Swiss reference level for refurbished residential buildings is exceeded by 11 %.

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 23 % to the CED and holds the Swiss reference value.

Conclusion: The primary energy demand (non renewable) of the refurbished residential property meets the target value. The main impact is caused by the operation phase.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the residential building E per m² energy reference area and 60 years lifespan.

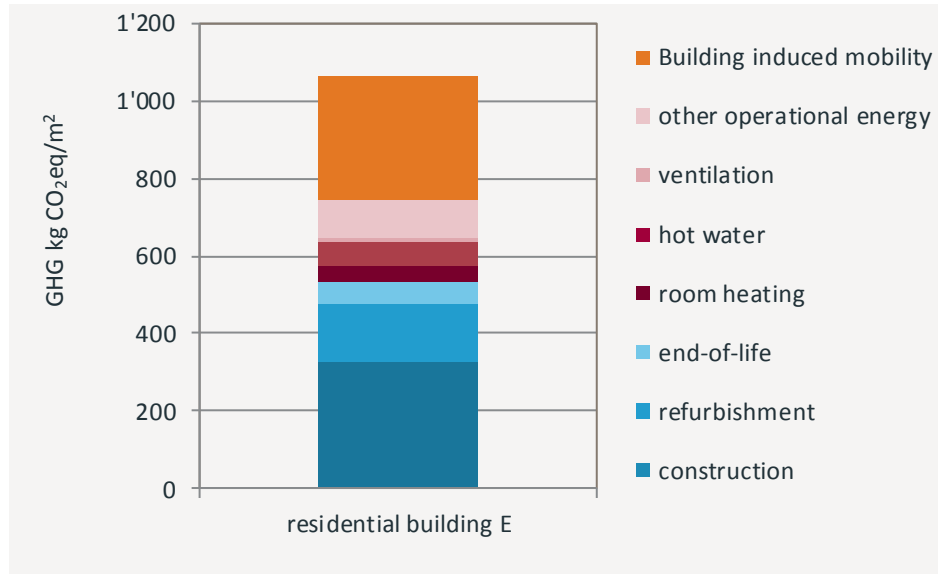


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the residential building E .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 50 %. Within the construction phase the materialization is dominating the GWP (61 %). The main impacts come from the ceilings, windows as well as from the infrastructure. The renewal per year contributes about 28 % to the GWP of the total construction. The deconstruction has a share of 11 %. The reference level regarding GWP for new residential buildings is slightly exceeded.

Operation: In the building's use phase (20 %) the GWP is mainly influenced by electricity demand (46 %), followed by the hot water provision (30 %). The residential building E is heated with a borehole heat exchanger. The reference level is exceeded by 40 %.

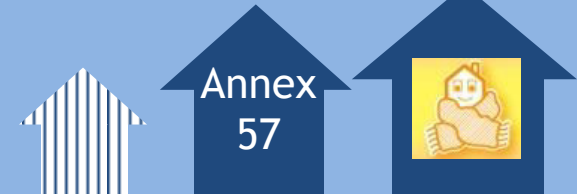
Induced mobility: The global warming potential of the induced mobility has a share of 30 % on the total GHG emissions and meets the reference value.

Conclusion: The impacts of the construction stage on the GWP are most dominant. The construction itself dominates the results. Overall the GWP of the residential building E slightly exceeds the target level for new residential buildings.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Residential E, new construction
Energy-standard	net-positive
Gross floor area/ Net floor area	2'487 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 2'156 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, heat distribution with floor heating Cooling: n/a
Final energy demand electricity	Ventilation 6 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 39 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 57 MJ/m ² a (per energy reference area) Hot water 50 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	-
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

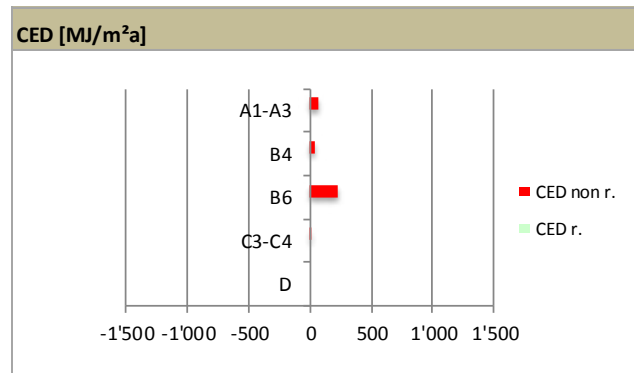
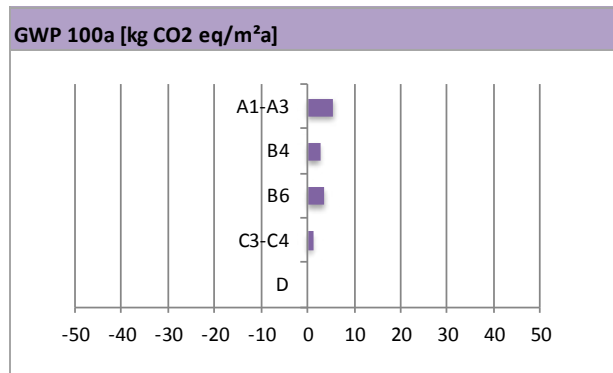
Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Building pit Backfill Foundation plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH10	Residential building E	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	5.44E+00	2.51E+00	3.49E+00	9.86E-01	0.00E+00
CED non r.	MJ/m ² a	6.86E+01	4.09E+01	2.23E+02	5.73E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



2.11 Retirement home A



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The retirement home A (refurbished building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The retirement home A has a total env. impact of 22'000 eco-points/m²a, primary energy demand (non renew.) of 390 MJ/m²a and a GWP of 17 kg CO₂/m²a. All indicators are dominated by the use stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This retirement home A is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: Retirement home

Size: 10'532 m² gross floor area, 9'843 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed in 2011



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Retirement home A

The property was built in 1976 and serves multifunctional purposes: retirement home, social center, coffee shops, shops and public parking. The new concept still encompasses all these functions. The retirement home was refurbished completely. Today there are 120 apartments of various size. The pensioners have different social services to make demand of.

All buildings are energetically refurbished and fulfill now the criteria of Minergie. Heating is covered by district heat.

CHARACTERISTIC FACTORS OPERATION

Floor area	10'532 m ²
Energy reference area	9'843 m ²
Energy demand room heating	68 MJ/m ² a
Energy demand hot water	50 MJm ² a
Energy demand electrical power	48 MJ/m ² a
- Energy demand ventilation:	10 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade A
Distance for groceries	0.1 km
Parking spots	0.07 parking spots per pensioner and employee
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)



DETAILED RESULTS OF THE RETIREMENT HOME A, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the retirement home A in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	backfill	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	fundament	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	ceiling	45	2'713	2'137	168	408	0.3	19.8	14.2	2.4	3.2	0.03	1.93	1.66	0.09	0.18
	roof	735	44'084	12'005	12'005	20'075	15.6	935.0	460.0	460.0	15.0	0.96	57.73	14.28	14.28	29.16
	pillars	2	137	119	-	19	0.0	0.9	0.7	-	0.1	0.00	0.12	0.11	-	0.01
	outer walls basement	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	outer walls upper floors	124	7'461	4'344	1'752	1'364	1.2	71.9	41.9	23.6	6.4	0.14	8.56	5.18	2.39	0.99
	windows	509	30'550	13'178	13'178	4'194	5.9	353.8	175.4	175.4	3.1	0.40	24.11	11.18	11.18	1.74
	inner walls raw	132	7'926	6'801	-	1'125	1.1	68.6	60.0	-	8.5	0.13	7.51	7.06	-	0.45
	separation walls/inner doors	680	40'815	18'878	18'878	3'060	9.7	584.1	285.7	285.7	12.7	0.64	38.50	18.78	18.78	0.95
	flooring	191	11'444	4'260	4'260	2'923	2.4	141.3	68.9	68.9	3.4	0.15	9.06	3.27	3.27	2.52
	wall cover	754	45'249	14'855	21'892	8'502	6.7	402.0	134.7	256.3	11.1	0.51	30.61	8.24	13.36	9.01
	ceiling cover	96	5'775	2'168	2'744	862	1.3	77.6	32.8	42.7	2.1	0.08	4.99	1.89	2.31	0.79
	infrastructure	3'433	205'985	97'798	101'098	7'089	22.9	1'376.9	668.8	695.1	13.0	1.46	87.73	38.10	39.70	9.93
	sum building	6'702	402'139	176'542	175'975	49'623	67.2	4'031.9	1'943.2	2'010.2	78.5	4.51	270.84	109.75	105.36	55.73
Operation	room heating	1'770	106'193				56.8	3'405.6				3.18	190.68			
	hot water	1'902	114'123				61.0	3'659.9				3.42	204.92			
	ventilation	1'252	75'098				26.3	1'580.8				0.41	24.73			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	9'680	580'780				244.2	14'653.2				8.57	514.32			
Building induced mobility	sum mobility	5'366	321'970				82.0	4'920.0				4.17	250.00			
sum total	construction, operation und building induced mobility	21'748					393.4					17.25				
target							-					-				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the retirement home A per m² energy reference area and 60 years lifespan.

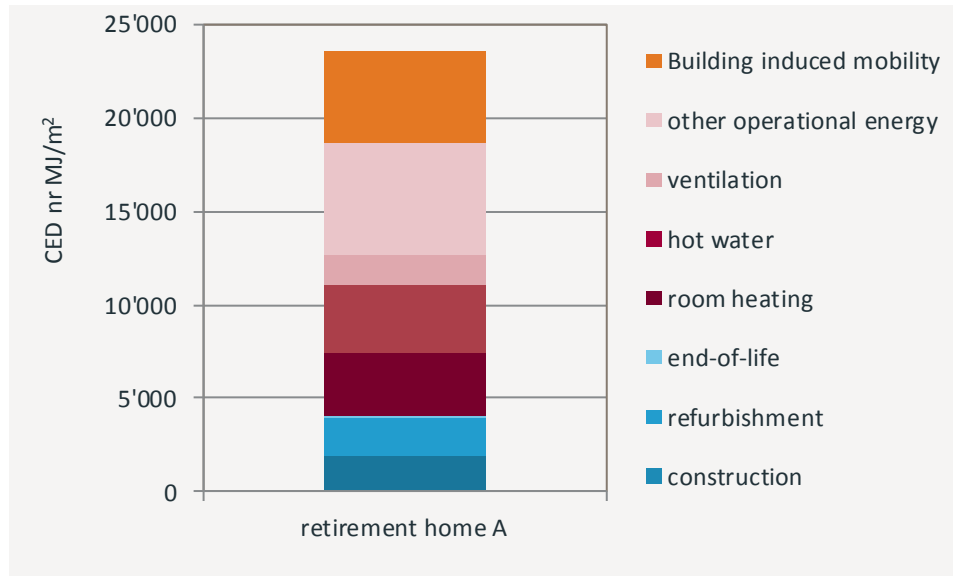


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the retirement home A .

Total construction (construction, renewal and deconstruction): The total construction has a share of 17 % to the overall primary energy demand (non renew.).

The construction itself contributes to 48 % to this. The main impacts come from the roof, the windows, the walls, the wall covers as well as from the infrastructure. The renewal phase contributes to 50 % to the primary energy demand of the total construction. In comparison the deconstruction has an impact of 2 %.

Operation: With 62 % the total operation has the main impact on the primary energy demand. The largest impact is caused by the electricity demand (41 %), followed by the hot water provision (25 %).

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 21 % to the CED.

Conclusion: The primary energy demand (non renewable) of the retirement home A is dominated by the use stage.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the retirement home A per m² energy reference area and 60 years lifespan.

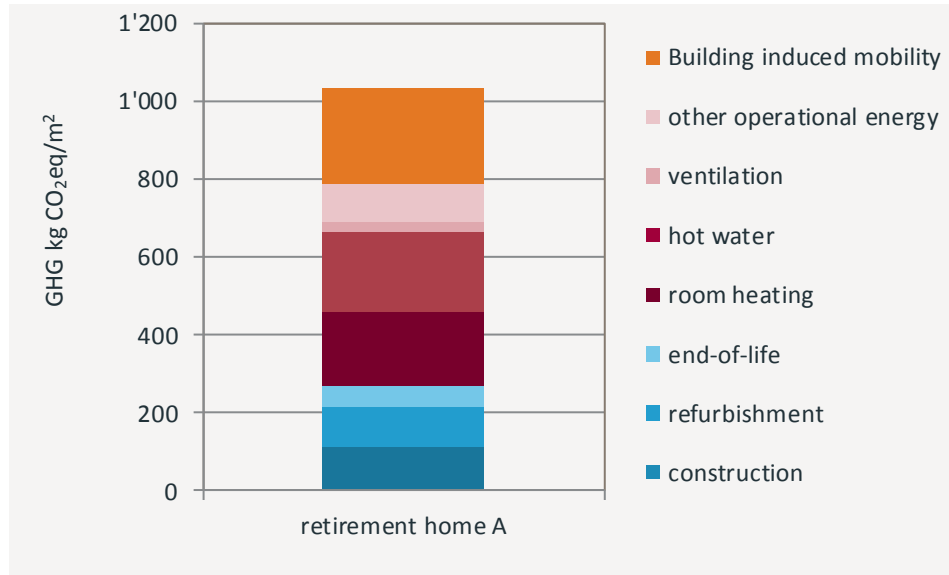


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the retirement home A .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 26 %. Within the construction phase the materialization is dominating the GWP (41 %). The main impacts come from the roof, the inner walls, windows as well as from the infrastructure. The renewal per year contributes about 39 % to the GWP of the total construction. The deconstruction has a share of 21 %.

Operation: In the building's use phase (50 %) the GWP is mainly influenced by the hot water provision (40 %), followed by the room heating (37 %). The retirement home A is heated with a district heat.

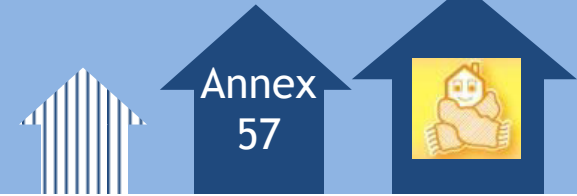
Induced mobility: The global warming potential of the induced mobility has a share of 24 % on the total GHG emissions.

Conclusion: The impacts of the use stage on the GWP are most dominant. The hot water provision dominates the results.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Retirement home A, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	10'532 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 9'843 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: district heat, heat distribution with floor heating Cooling: n/a
Final energy demand electricity	Ventilation 10 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 68 MJ/m ² a (per energy reference area) Hot water 50 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered

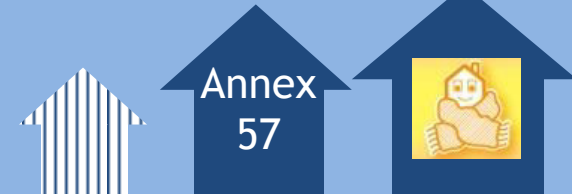


MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Ceilings Roof Pillars External walls (above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

DOCUMENTATION REQUIREMENTS

RETIREMENT HOME A (III)

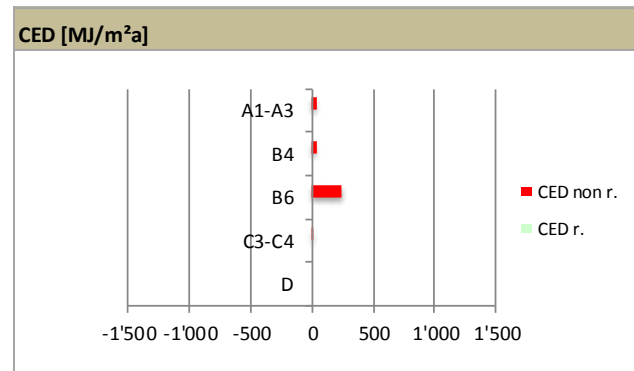
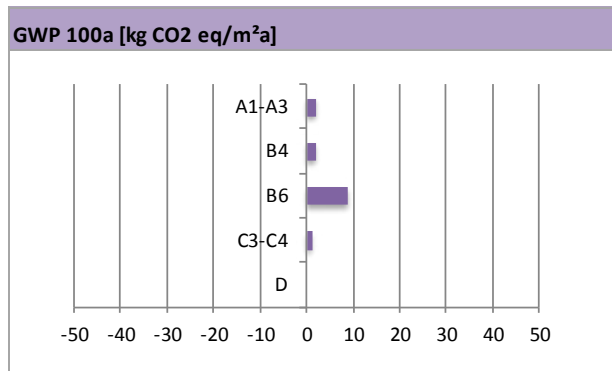


MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH11	Retirement home A	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	1.83E+00	1.76E+00	8.57E+00	9.29E-01	0.00E+00
CED non r.	MJ/m ² a	3.24E+01	3.35E+01	2.44E+02	1.31E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)



2.12 Retirement home B



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The rest home B (refurbished building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The rest home B has a total env. impact of 31'000 eco-points/m²a, primary energy demand (non renew.) of 470 MJ/m²a and a GWP of 17 kg CO₂/m²a. The indicator CED is dominated by the use stage, while the GWP is mostly influenced by the construction stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building and the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This rest home B is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

BUILDING KEY FACTS

Intended use: Retirement/Rest home

Size: 14'479 m² gross floor area, 11'186 m² energy reference area

Location: Zurich, Switzerland

Building year: Completed in 2010



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

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Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Rest home B

The rest home B was constructed in 1983 and it corresponds to the zeitgeist of the 70ies. In encompassing work the building was renewed optically and energetically. The façade, the roof, the interior and the technical equipment was refurbished. The building consists of six upper floors, ground floor and basement and a green flat roof.

The buildings meet the requirements of Minergie standard. Hot water and room heating are covered by an electrical heat pump equipped with a borehole heat exchanger and floor heating. Peak demand is covered with a gas-fueled boiler. The rest home has automatic ventilation.

CHARACTERISTIC FACTORS OPERATION

Floor area	14'479 m ²
Energy reference area	11'186 m ²
Energy demand room heating	98 MJ/m ² a
- Heat pump	95 %
- Gas	5 %
Energy demand hot water	50 MJm ² a
- Heat pump	80 %
- Gas	20 %
Energy demand electrical power	53 MJ/m ² a
- Energy demand ventilation:	15 MJ/m ² a
- Energy demand residual operation:	38 MJ/m ² a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade B
Distance for groceries	0.8 km
Parking spots	0.06 parking spots per pensioner and employee
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)



DETAILED RESULTS OF THE REST HOME B, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the rest home B in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	2.44	146	146	-	-	0.0	1.8	1.8	0.0	0.0	0.00	0.12	0.12	-	-
	backfill	1	77	76	-	1	0.0	0.9	0.9	0.0	0.0	0.00	0.06	0.06	-	0.00
	fundament	177	10'620	8'897	-	1'723	1.2	69.1	57.2	0.0	11.9	0.11	6.77	5.93	-	0.84
	ceiling	1'033	61'978	53'174	2'001	6'804	6.5	391.3	321.8	18.2	51.3	0.70	41.91	37.97	1.16	2.77
	roof	838	50'257	24'222	7'488	18'547	9.0	541.2	345.1	148.0	48.1	0.61	36.57	11.64	4.37	20.56
	pillars	41	2'442	2'413	-	29	0.2	13.2	13.0	0.0	0.2	0.01	0.86	0.85	-	0.01
	outer walls basement	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	-	-	-	-	-
	outer walls upper floors	1'773	106'401	64'023	41'386	993	13.9	832.7	481.4	344.6	6.7	1.02	61.26	38.61	22.29	0.36
	windows	611	36'639	16'507	16'507	3'625	7.8	465.9	231.4	231.4	3.1	0.53	31.66	15.05	15.05	1.56
	inner walls raw	182	10'914	9'359	-	1'555	2.8	166.4	155.5	0.0	10.9	0.25	14.86	14.30	-	0.57
	separation walls/inner doors	389	23'356	8'473	8'473	6'410	4.2	254.0	123.4	123.4	7.1	0.26	15.82	6.85	6.85	2.12
	flooring	942	56'497	23'284	23'284	9'930	6.9	414.3	184.6	184.6	45.1	0.84	50.10	23.32	23.32	3.46
	wall cover	1'050	63'014	27'256	27'256	8'502	8.4	504.2	243.6	243.6	16.9	0.67	40.25	17.53	17.53	5.19
	ceiling cover	512	30'734	14'646	14'646	1'441	3.7	221.1	108.8	108.8	3.5	0.21	12.84	5.89	5.89	1.07
	infrastructure	4'143	248'592	118'567	122'936	7'089	27.8	1'667.7	793.1	861.7	13.0	1.74	104.50	45.38	49.19	9.93
	sum building	11'694	701'669	371'043	263'977	66'649	92.4	5'543.8	3'061.6	2'264.3	217.9	6.96	417.58	223.50	145.65	48.43
Operation	room heating	3'902	234'093				83.3	4'996.3				1.59	95.43			
	hot water	2'836	170'144				66.6	3'996.5				1.86	111.30			
	ventilation	1'877	112'645				39.5	2'371.1				0.62	37.10			
	other operational energy	4'756	285'366				100.1	6'006.9				1.57	93.98			
	sum operation	13'371	802'248				289.5	17'370.8				5.63	337.81			
Building induced mobility	sum mobility	5'733	344'000				87.7	5'260.0				4.50	270.00			
sum total	construction, operation und building induced mobility	30'799					469.6					17.09				
target							-					-				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the rest home B per m² energy reference area and 60 years lifespan.

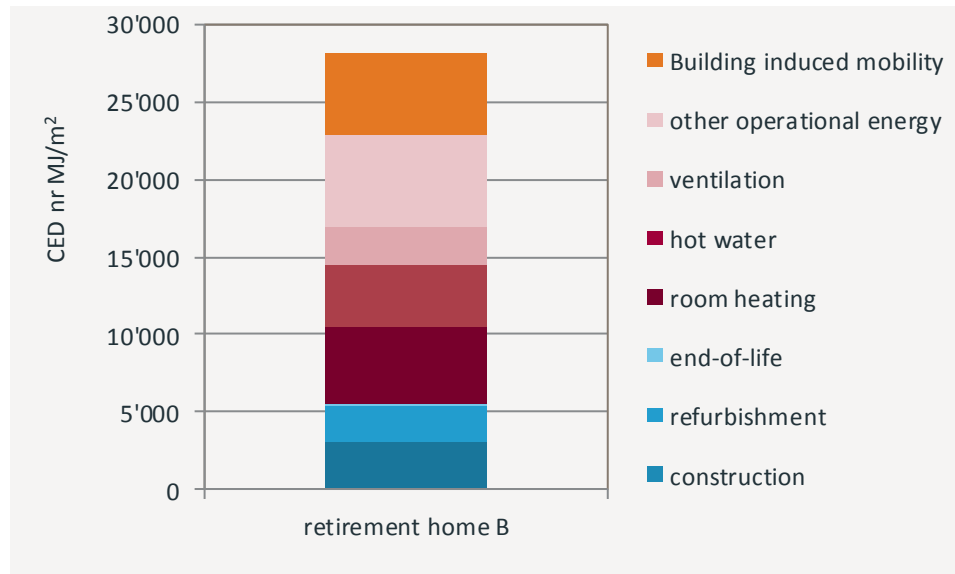


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the rest home B .

Total construction (construction, renewal and deconstruction): The total construction has a share of 20 % to the overall primary energy demand (non renew.).

The construction itself contributes to 55 % to this. The main impacts come from the roof, ceiling, inner walls, windows, wall covers, flooring as well as from the infrastructure. The renewal phase contributes to 41 % to the primary energy demand of the total construction. In comparison the deconstruction has an impact of 4 %.

Operation: With 62 % the total operation has the main impact on the primary energy demand. The largest impact is caused by the electricity demand (35 %), followed by the room heating (29 %).

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 19 % to the CED.

Conclusion: The primary energy demand (non renewable) of the rest home B is dominated by the use stage, especially the operational electrical energy.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the rest home B per m² energy reference area and 60 years lifespan.

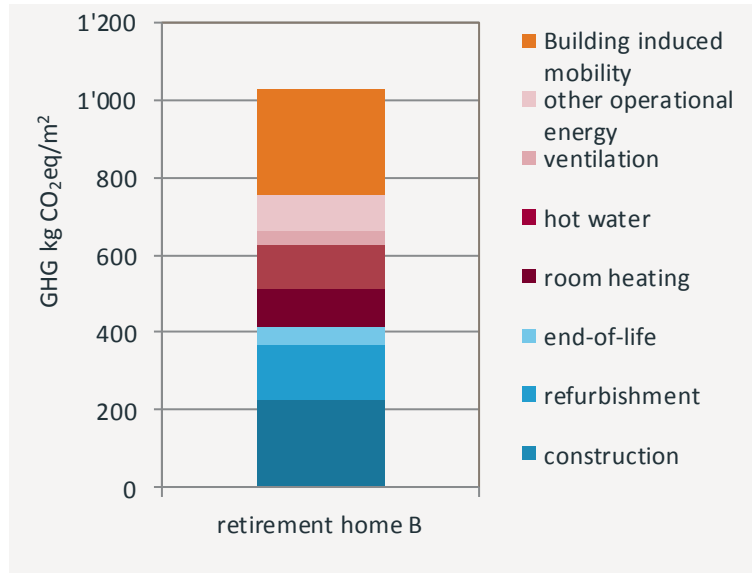


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the rest home B .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 41 %. Within the construction phase the materialization is dominating the GWP (54 %). The main impacts come from the roof, ceiling, the inner walls, windows, wall covers, flooring as well as from the infrastructure. The renewal per year contributes about 35 % to the GWP of the total construction. The deconstruction has a share of 12 %.

Operation: In the building's use phase (33 %) the GWP is mainly influenced by the hot water provision (33 %), followed by the room heating and electricity demand (28 % each). The rest home B is heated with an electric heat pump equipped with a borehole heat exchanger. Peak demand is covered with a gas-fueled boiler.

Induced mobility: The global warming potential of the induced mobility has a share of 26 % on the total GHG emissions.

Conclusion: The impacts of the construction stage on the GWP are most dominant.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Retirement home B, refurbishment
Energy-standard	net-positive
Gross floor area/ Net floor area	14'479 m ²
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 11'186 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, peak demand is covered with a gas-fueled boiler, heat distribution with floor heating Cooling: n/a
Final energy demand electricity	Ventilation 15 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 38 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 98 MJ/m ² a (per energy reference area) - Heat pump 95 % - Gas 5 % Hot water 50 MJ/m ² a (per energy reference area) - Heat pump 80 % - Gas 20 %
Final energy demand for cooling	0 MJ/m ² a
Benchmark	-
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life, operation and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Only the materials used for the refurbishment were considered. Construction pit Backfilling Fundament plate Ceilings Roof Pillars External walls (above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)

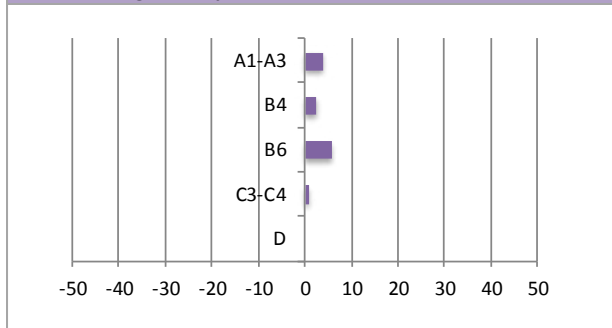
MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH12	Retirement home B	

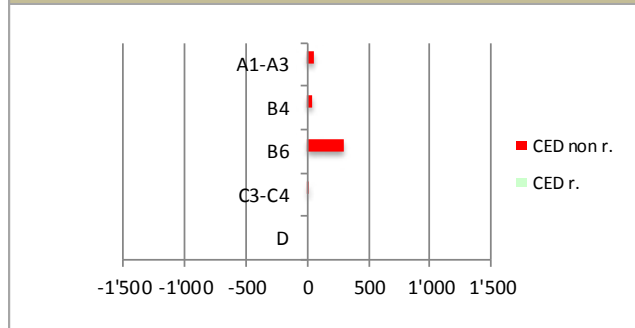
Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	3.72E+00	2.43E+00	5.63E+00	8.07E-01	0.00E+00
CED non r.	MJ/m ² a	5.10E+01	3.77E+01	2.90E+02	3.63E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)

GWP 100a [kg CO₂ eq/m²a]



CED [MJ/m²a]



2.13 Retirement home C



Case study

The total environmental impacts of buildings

KEY OBSERVATIONS

The retirement home C (refurbished building) was analyzed in terms of construction and operation of the buildings as well as the induced mobility. The env. impacts were assessed as total env. impact, non renewable primary energy demand and global warming potential (GWP). The latter two are shown in detail.

The retirement home C has a total env. impact of 29'000 eco-points/m²a, primary energy demand (non renew.) of 380 MJ/m²a and a GWP of 17 kg CO₂/m²a. The indicator CED is dominated by the use stage, while the GWP is mostly influenced by the construction stage.

This example shows that the topic of env. impacts of buildings is not only very important but also very complex. The environmental impact depends largely on the materialization, the usage of the building. End the type of energy source for the provision of heat and hot water.

OBJECTIVES OF CASE STUDY

The main target is to perform a Life Cycle Assessment (LCA) to evaluate the total environmental (env.) impact of the building regarding construction, operation, end-of-life and induced mobility. This assessment is performed in the context of the discussion about reference and target values for env. impacts of buildings. 33 buildings located in Zurich, Switzerland, were analyzed. This retirement home C is one of the sample and is presented here. The study evaluates:

- The influence of the different life cycle stages: construction, renewal and deconstruction at the end-of-life
- The importance of the annual operational energy demand: heating, hot water, ventilation, and residual operational energy demand
- The significance of the induced mobility
- The impacts related to different building parts
- The impacts related to different building materials

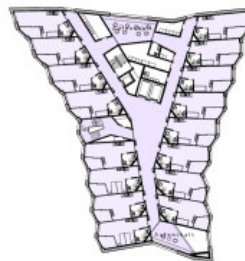
BUILDING KEY FACTS

Intended use: Retirement home

Size: Gross floor area n/a, energy reference area 8'745 m²

Location: Zurich, Switzerland

Building year: Not completed yet



ABBREVIATIONS

CED cumulative energy demand

GHG greenhouse gases

GWP global warming potential

LCA life cycle assessment

nr non renewable



Building life cycle stages included in the study, according to EN15978

A 1-3 Product stage			A 4-5 Construction process stage		B 1-7 Use stage							C 1-4 End-of-Life				D Next product system
Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction/demolition	Transport to EoL	Waste processing	Disposal	Reuse, recovery or recycling potential
X	X	X			X			X		X				X	X	X

LCA BACKGROUND

Average life time of buildings: 60 years
 Calculation of total env. impact: Ecological scarcity 2006 (Frischknecht et al. 2008)
 Calculation of Energy: Cumulative energy demand, differing non-renewable and renewable primary energy (Frischknecht et al. 2007)
 Calculation of GWP: GWP 100 years (IPCC 2007, TS 2)
 Databases used: ecoinvent data v2.2

REFERENCES

- Wyss et al. (2014) Zielwert Gesamtumweltbelastung Gebäude, Bundesamt für Energie (BfE), Bundesamt für Umwelt (Bafu), Stadt Zürich Amt für Hochbauten
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- SIA D 0236 (2011), SIA-Effizienzpfad Energie – Ergänzungen und Fallbeispiele zum Merkblatt SIA 2040

Production and construction stage modeling: All impacts from the raw material extraction and the manufacturing of the building materials are included. Transport to the building site is not considered, neither are the impacts of the construction phase itself. The datasets base on data of the Swiss public coordination conference of construction and property services of the public building owners (KBOB).

Operation stage modeling: In the building's use phase the energy consumption of space heating, heat distribution, hot water generation as well as ventilation are considered.

The replacements of building materials and components during the service life of the building are calculated based on the average lifetime of the component and the building; p. ex. a certain material with a life time of 30 years is accounted for twice because it is installed two times in the building's service life of 60 years.

End of life stage and next product system modeling: The EoL is modelled according to the current average Swiss disposal routes. Recycled materials cause no waste management impacts, nor are any credits accounted for. Other materials are landfilled or incinerated. The environmental impacts caused by waste management are accounted for.



Retirement home C

In Zurich-Wipkingen a new modern retirement home is planned. The existing retirement home does not meet the requirements of modern living no more. The new building shall fulfill the criteria of Minergie-P.

CHARACTERISTIC FACTORS OPERATION

Floor area	n/a
Energy reference area	9'843 m²
Energy demand room heating	20 MJ/m²a
Energy demand hot water	50 MJm²a
Energy demand electrical power	45 MJ/m²a
- Energy demand ventilation:	6 MJ/m²a
- Energy demand residual operation:	39 MJ/m²a

CHARACTERISTIC FACTORS INDUCED MOBILITY

Type	City center
Public transport	grade B
Distance for groceries	0.7 km
Parking spots	0.06 parking spots per pensioner and employee
Public transport subscriptions	0.25 permanent public transport subscriptions (Swiss average)



DETAILED RESULTS OF THE RETIREMENT HOME C, ZURICH

Total environmental impact, non renewable primary energy demand and global warming potential of the retirement home C in Zurich, referring to 1 m² energy reference area and 60 years service life.

	indicator	sum environmental impact					primary energy demand non renewable					greenhouse gas emissions				
	unit	UBP/m ² a	UBP/m ²				MJ/m ² a	MJ/m ²				kg CO ₂ /m ² a	kg CO ₂ /m ²			
	EKG-number	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life	sum amortized	sum	construction	refurbishment	end-of-life
Building's construction	construction pit	21.99	1'319	1'319	-	-	0.3	15.9	15.9	0.0	0.0	0.02	1.06	1.06	-	-
	backfill	5	327	327	-	-	0.1	4.0	4.0	0.0	0.0	0.00	0.26	0.26	-	-
	fundament	1'197	71'842	58'778	-	13'064	7.8	465.1	378.8	0.0	86.3	0.87	52.42	45.54	-	6.88
	ceiling	2'474	148'469	126'834	332	21'303	17.2	1'030.5	871.4	4.8	154.2	1.79	107.16	96.95	0.22	9.99
	roof	1'287	77'210	55'678	9'611	11'920	11.5	688.6	500.2	141.2	47.2	0.98	58.84	41.07	6.39	11.38
	pillars	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	-	-	-	-	-
	outer walls basement	615	36'903	27'179	2'022	7'702	5.1	305.4	229.5	45.0	30.9	0.60	35.71	25.44	2.99	7.28
	outer walls upper floors	1'433	85'982	61'568	19'774	4'640	11.2	670.9	467.2	167.2	36.4	0.89	53.13	40.34	10.91	1.88
	windows	645	38'680	16'689	16'689	5'302	7.5	452.5	224.4	224.4	3.8	0.51	30.82	14.29	14.29	2.24
	inner walls raw	1'064	63'844	54'587	-	9'257	6.7	404.3	333.8	0.0	70.5	0.83	50.02	46.24	-	3.78
	separation walls/inner doors	891	53'480	24'046	24'046	5'387	10.0	600.9	294.1	294.1	12.6	0.57	34.42	16.26	16.26	1.90
	flooring	245	14'728	5'582	4'055	5'092	3.2	194.7	122.5	69.1	3.0	0.24	14.25	4.74	3.60	5.92
	wall cover	385	23'108	10'841	10'841	1'425	2.7	161.4	78.9	78.9	3.7	0.16	9.48	4.17	4.17	1.14
	ceiling cover	177	10'609	4'282	4'668	1'659	2.2	133.9	62.4	69.2	2.3	0.16	9.43	3.84	4.12	1.48
	infrastructure	3'433	205'985	97'798	101'098	7'089	22.9	1'376.9	668.8	695.1	13.0	1.46	87.73	38.10	39.70	9.93
	sum building	13'875	832'485	545'509	193'136	93'840	108.4	6'504.8	4'251.9	1'789.1	463.8	9.08	544.74	378.31	102.65	63.79
Operation	room heating	861	51'649				6.5	391.5				0.33	19.76			
	hot water	2'875	172'475				59.9	3'592.7				0.94	56.21			
	ventilation	751	45'058				15.8	948.5				0.25	14.84			
	other operational energy	4'881	292'877				102.7	6'165.0				1.61	96.45			
	sum operation	9'368	562'060				185.0	11'097.6				3.12	187.26			
Building induced mobility	sum mobility	5'709	342'510				87.3	5'240.0				4.50	270.00			
sum total	construction, operation und building induced mobility	28'951					380.7					16.70				
target							-					-				



PRIMARY ENERGY DEMAND, NON RENEWABLE

Primary energy demand, non renewable, from the construction, operation and induced mobility for the retirement home C per m² energy reference area and 60 years lifespan.

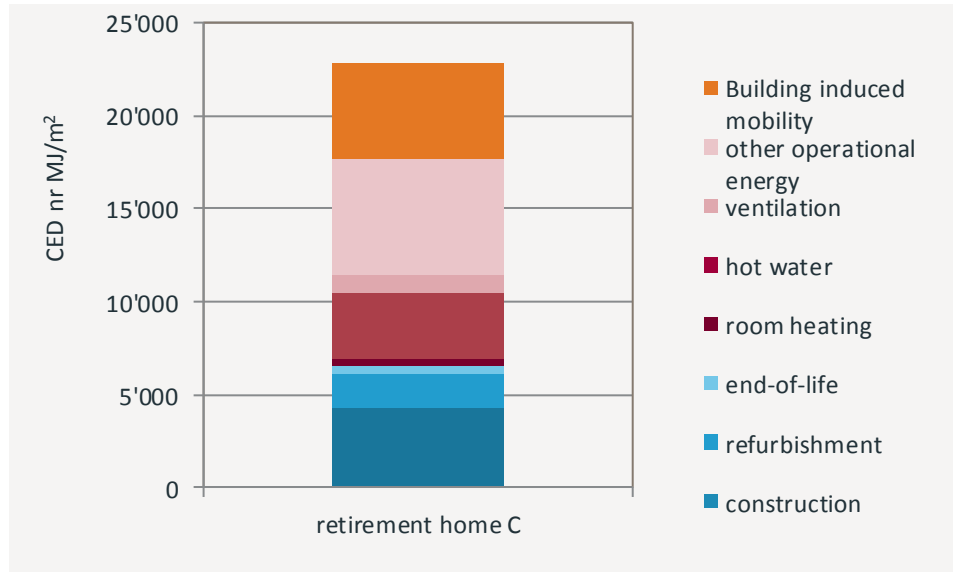


Figure 1: Contribution to the primary energy demand, non renewable, from the construction, operation and induced mobility for the retirement home C .

Total construction (construction, renewal and deconstruction): The total construction has a share of 28 % to the overall primary energy demand (non renew.).

The construction itself contributes to 65 % to this. The main impacts come from the fundament, ceilings, the roof, the windows, the walls as well as from the infrastructure. The renewal phase contributes to 28 % to the primary energy demand of the total construction. In comparison the deconstruction has an impact of 7 %.

Operation: With 49 % the total operation has the main impact on the primary energy demand. The largest impact is caused by the electricity demand (56 %), followed by the hot water provision (32 %).

Induced mobility: The primary energy demand (non renew.) of the induced mobility has a share of 23 % to the CED.

Conclusion: The primary energy demand (non renewable) of the retirement home C is dominated by the use stage, especially the electricity demand.



GLOBAL WARMING POTENTIAL (GWP)

Global warming potential from the construction, operation and induced mobility for the retirement home C per m² energy reference area and 60 years lifespan.

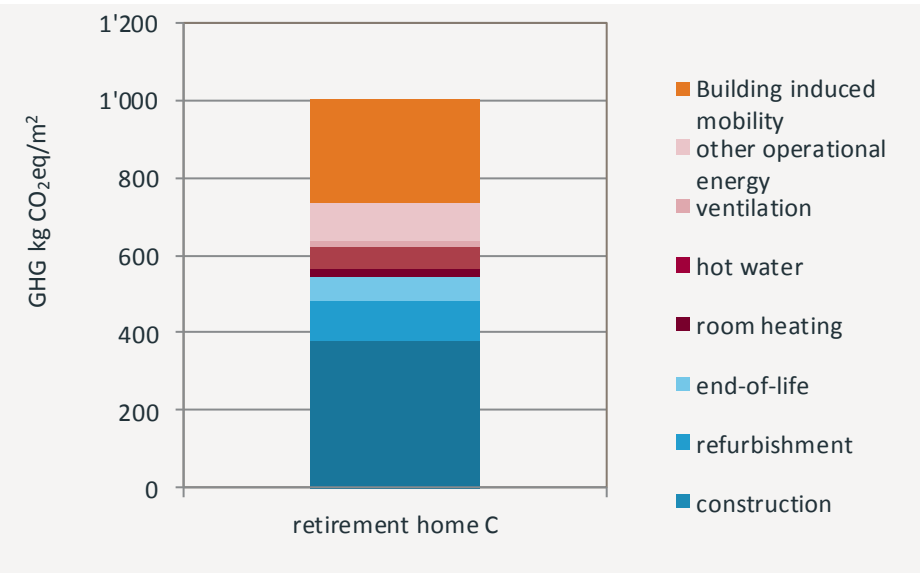


Figure 2: Contribution to the global warming potential from the construction, operation and induced mobility for the retirement home C .

Total construction (construction, renewal and deconstruction): The construction phase contributes to the total global warming potential by 54 %. Within the construction phase the materialization is dominating the GWP (69 %). The main impacts come from the fundament, ceilings, the roof, walls as well as from the infrastructure. The renewal per year contributes about 19 % to the GWP of the total construction. The deconstruction has a share of 12 %.

Operation: In the building's use phase (19 %) the GWP is mainly influenced by the electricity demand (52 %) followed by the hot water provision (30 %). The retirement home C is heated with a heat pump.

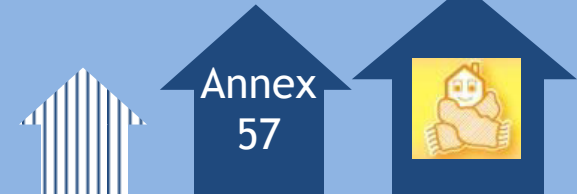
Induced mobility: The global warming potential of the induced mobility has a share of 27 % on the total GHG emissions.

Conclusion: The impacts of the construction stage on the GWP are most dominant. The construction itself dominates the results.



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Location /climate and or heating degree days / cooling?	Switzerland / moderate climate
Building/ Usage type	Residential home – Retirement home C, new construction
Energy-standard	net-positive
Gross floor area/ Net floor area	n/a
Gross volume/ Net volume	n/a
Reference area for EE/EC	energy reference area 8'745 m ²
Surface/Volume ratio (m-1)	n/a
Construction method	Massive construction (concrete, brick)
Thermal insulation	Insulation of walls, roof insulation
Ventilation system	Automatic ventilation
Heating and cooling system	Heating: electric heat pump equipped with a borehole heat exchanger, heat distribution with floor heating Cooling: n/a
Final energy demand electricity	Ventilation 6 MJ/m ² a (per energy reference area) Appliances, lighting, services, etc. 39 MJ/m ² a (per energy reference area)
Final energy demand for heating and hot water	Room heating 20 MJ/m ² a (per energy reference area) Hot water 50 MJ/m ² a (per energy reference area)
Final energy demand for cooling	0 MJ/m ² a
Benchmark	Minergie-P-eco
Purpose of assessment	to determine CED and GHG emissions as well as total environmental impact for construction, use, end-of-life and the induced mobility of the building
Assessment methodology	According to the methodology of ecoinvent and to SIA 2032 guidance
Reference Study Period	60 years
Included life cycle stages	From cradle to grave - Construction stage - use stage - end-of-life stage - induces mobility No benefits for potential recycling were considered



MINIMUM DOCUMENTATION REQUIREMENTS

Parameter	Case Study Description/ Minimum Documentation Requirements (Type 1-4)
Included parts of the building	Construction pit Backfilling Fundament plate Ceilings Roof Pillars External walls (underground and above ground) Windows, doors Internal walls Internal doors, dividing walls Flooring Wall covers Ceiling covers Infrastructure (ventilation, heating, sanitary equipment, electrical equipment)
Scenarios and assumptions used	Recycling at the end-of-life
Accounting of electricity mix	static emissions factors, Swiss consumer mix
Databases used	Ecoinvent v2.2 and v2.2+
LCA Software used	Simapro 7.3.3
Method of materials quantification	LCI
Values and sources of primary energy and emission factors	KBOB-recommendation (www.kbob.ch) ecoinvent
Character of the indicator used	Optimum or modified (e.g. tables 9, 11 or 13 in ST1 draft report)
Indicators assessed	CED non renewable (according to Frischknecht et al, 2007) GHG emissions (according to IPCC 2007 and 2013) total environmental impact (according to the method of ecological scarcity 2006 and 2013)



MINIMUM DOCUMENTATION REQUIREMENTS - RESULTS

Life Cycle Assessment		57 Annex 57
CH13	Retirement home C	

Indicator	Unit	Product stage	Use stage		End-of-Life	Next product system
		A1-A3: Product stage	B4: Replacement	B6: Operational energy use	C3-C4: End-of-Life	D: Reuse, recovery or recycling potential
GWP	kgCO ₂ /m ² a	6.31E+00	1.71E+00	3.12E+00	1.06E+00	0.00E+00
CED non r.	MJ/m ² a	7.09E+01	2.98E+01	1.85E+02	7.73E+00	0.00E+00
CED r.	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
CED complete	MJ/m ² a	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

* GWP (Global Warming Potential for a 100-year time horizon)

